

Misallocation, Variable Markups, and the Labour Share: Plant-Level Evidence from Indian Manufacturing, 2012–2022*

Laxmikanta Sarangi Falguni Pattanaik

October 2026

Working paper

Abstract

How much output does Indian manufacturing lose to misallocation, and do variable markups change the answer? On a new panel of about 39,000 organised-sector plants per year for 2012–2022, we measure within-industry misallocation with the Hsieh–Klenow accounting and inside a calibrated Kimball demand system. Equalising marginal revenue products within industries would raise measured TFP by 121–184 per cent among plants holding half of value added, a static bound, not a policy gain. Capital wedges exceed labour wedges in every year and state, and the wedges are persistent rather than transitory noise. Variable markups explain under two per cent of dispersion yet change the gain by 7–19 per cent, in a direction set by the demand system: two of four alternatives reverse it. Equalisation lowers the markup-implied labour share by 2.65 percentage points (0.94 under domestic capital elasticities). Wages are held fixed, so this prices the markup margin, not workers’ incomes.

JEL codes: O11, O47, D24, D61, L11, E25.

Keywords: misallocation; TFPR dispersion; Kimball demand; variable markups; labour share; Indian manufacturing; Annual Survey of Industries.

*Department of Humanities and Social Sciences, Indian Institute of Technology Roorkee. This paper is the second of a series based on a newly constructed plant panel from the Annual Survey of Industries; the companion papers measure markups and markdowns (working paper 1) and their macroeconomic determinants (working paper 3). ASI unit records are used under licence from the Ministry of Statistics and Programme Implementation and cannot be redistributed; all aggregate results and code are available from the authors. All errors are ours.

Contents

1	Introduction	8
2	Related literature	13
2.1	Two traditions: the covariance and the wedge	13
2.2	The Hsieh–Klenow mapping, in four steps	14
2.3	The critiques that define the frontier	16
2.4	Variable markups: the critique that is also a mechanism	17
2.5	The empirical record, and the India strand	17
2.6	Indian antecedents, and the closest one	19
2.7	Where this paper sits, and what it adds	20
3	Data	21
3.1	Source and sample	21
3.2	Cleaning and the estimation sample	23
3.3	Parameters taken from outside the data	23
4	Framework	24
4.1	Technology, wedges, and revenue productivity	24
4.2	From CES to Kimball: letting the markup vary	26
4.3	Identifying the two demand parameters	27
4.4	Four equilibria and what their differences isolate	31
4.5	Size-dependent wedges: the HKKR diagnostics	32
5	Results	32
5.1	The anatomy of dispersion	32
5.2	How much of the measured problem is the accounting?	34
5.3	How much of the measured dispersion is transitory noise?	36
5.4	Who bears the wedges, and where the improvement came from	40
5.5	Calibration by year	42
5.6	Counterfactual gains: level, path, and the markup tax	44
5.7	Whose markups? A three-way triangulation	47
5.8	The labour-share counterfactual: what reallocation does to the markup- implied labour share	48
5.9	The size dimension: HKKR diagnostics on the organised sector	49
5.10	Geography: states, regions, and the rural–urban margin	51
5.11	Does labour-market power sit where labour is misallocated?	55

6	Robustness and sensitivity	59
6.1	Domestic capital elasticities: a degeneracy worth reporting	59
6.2	Book-value capital and the vintage channel	59
6.3	The demand system	62
6.4	Inference: what the estimates are worth	63
6.5	Accounting, demand system, and elasticities: where they are treated	65
6.6	Survey weights	65
6.7	The 2019 boundary case and other checks	66
7	Interpretation: market power and misallocation, jointly read	66
8	Conclusion	68
A	Sample construction and deviations from the reference implementation	74
B	Numerical implementation	75
C	The full state panel	75
D	Gross-output versus value-added accounting: detail	77

List of Tables

1	Findings at a glance (2012–2022 unless noted; “gain” = counterfactual TFP gain from within-industry marginal-product equalisation).	13
2	This paper against Liang (2023), whose framework it adopts. The two studies share an estimator and a data source and differ in window, calibration inputs and the questions asked of the solved model. Liang’s entries are as cited elsewhere in this paper; cells we cannot state from the published account are marked n.r. The rows that matter for novelty are the last four: the objects this paper adds are the labour-share counterfactual, the plant-level triangulation of markups, the within/between decomposition over a decade, and the cross-paper factor-market join — none of which is available from a single-cross-section design.	18

3	Where this paper sits. The columns are the choices that determine whether two misallocation estimates are comparable: the named data source, the period, whether output is observed as revenue or as physical quantity, which wedges the study actually recovers, and the method used to recover them. The output-measure column matters because (6) shows that everything which moves price and not quantity is scored as a distortion when only revenue is observed. Cells marked n.r. are not reported in a form we can state confidently. HK = Hsieh–Klenow accounting; OP = Olley–Pakes decomposition; PF = production function; GE = general equilibrium; MRPK, MRPL = marginal revenue product of capital, of labour.	22
4	Instrumenting the market share in (17). Every column absorbs plant and year effects and clusters on the plant, exactly as the baseline does. “First-stage F ” is the Kleibergen–Paap Wald statistic. The instruments are individually overwhelming and jointly rejected, which is the point: they are not noisy readings of one latent share but measures of different notions of size.	31
5	Within-industry (NIC-4×year) dispersion of revenue productivity and factor marginal products, by year. Standard deviations of log deviations from industry aggregates; final column is $\text{var}(\log \text{TFPR}) \times 100$	33
6	Persistence of the within-industry wedges, 2012–2022. 429,534 plant-years on 132,196 plants; 255,385 one-year pairs and 107,713 five-year pairs. “Stay” is the probability that a plant in the top (or bottom) quintile of its industry-year wedge distribution is in the same quintile k years later; 0.20 is the value under independence. The permanent-share band runs from the five-year covariance ratio (a lower bound, since transitory shocks and classical error do not survive a five-year lag) to the between-plant variance share (an upper bound, since a plant mean over few years still carries noise).	38
7	Annual calibration targets: aggregate Lerner markup and implied sales-weighted demand-elasticity target. Comparison: Liang (2023), 2004: markup ≈ 1.52 , target 2.92.	43
8	Counterfactual TFP gains by reference year (per cent), each year recalibrated. Recalibration operates through the demand-elasticity parameter ς ; the superelasticity ratio ε/ς is pooled across years, following Liang (2023), so the year-to-year movement is not a full re-estimation of the demand system. “Market” lets markups re-optimize; “Planner” imposes marginal-cost pricing; “CES” freezes markups (HK benchmark). Final column: attainable share of the CES promise. 2019 is a boundary case (see text).	44

9	Sample flow from the cleaned panel to the counterfactual: plants the Kimball calibration cannot price. The solver of Liang (2023), used verbatim, drops any plant whose relative-output equation has no root in its search interval. Dropped shares of plants, employment and value added are of the cleaned panel for that year. The surviving N reproduces Table 8 exactly in every year. Source: <code>solver_dropcount.csv</code> , <code>39_solver_dropcount.do</code> .	45
10	Markups under three identification strategies, 2017 cross-section ($N = 37,699$; 33,497 matched to a production-side estimate). Source: <code>markup_three_methods_2017.csv</code> .	
11	The aggregate labour share under counterfactuals, 2017 reference year. Every change is measured from one common baseline, the observed allocation: the Kimball baseline reproduces the observed wage-bill share of value added in the estimation sample (0.3427) exactly. The CES model, solved on the same observed wedges, does not, and its misfit is shown as a row of its own. Source: <code>labor_share_counterfactual_ref2017.csv</code> ; observed share from <code>16_labor_share_counterfactual.do</code> .	49
12	Size-dependent wedges, year by year: share of within-industry TFPR variance explained by seven employment-size bins, and the size-neutralising counterfactual gain against the full-equalisation benchmark (both at $\sigma = 3$, HKKR machinery).	50
13	The TFPR size gradient, pooled 2012–2022: revenue-weighted regression of within-industry log-TFPR deviation on employment-size bins (base: smallest bin), NIC-4 $\times$ year effects. Bin 7 = 500+ workers.	50
14	HKKR’s motivating gradients inside the organised sector: wage premium by size bin (log points, relative to bin 1) and the white-collar (non-production) employment share.	51
15	Misallocation by region, pooled 2012–2022: within-industry variance ($\times 100$) of TFPR and MRPL; median markup and monopsony incidence from the companion paper for comparison.	52
16	Rural and urban misallocation by year: within-industry $\text{var}(\log \text{TFPR}) \times 100$.	52
17	Monopsony incidence (the companion paper) against factor-wedge dispersion (this paper), across the 21 states present in both, pooled 2012–2022. Pearson r with its two-sided p -value on 19 degrees of freedom, and Spearman ρ .	56
18	Each within-industry-year wedge regressed on the undepreciated fraction of the book capital stock. Estimation sample reproduced exactly (429,534 plant-years); the proxy is usable on the 237,507 with a positive gross stock. Source: <code>vintage_check.csv</code> , <code>vintage_deciles.csv</code> .	60

19	The capital wedge under book value and under perpetual-inventory capital. All wedges are within-industry-year, trimmed at the 1 per cent tails by year as elsewhere. 320,980 plant-years, the subset on which every capital variant is positive. The last row seeds the recursion on the gross rather than the net book stock. Source: <code>pim_wedges.csv</code> , <code>pim_persistence.csv</code>	61
20	2017 counterfactual gains under alternative demand systems (all recalibrated to the 2017 targets; no capped sectors). Comparison, Liang (2023) Table 5 (2004): translog 86.7, Dotsey–King 109.0, oligopoly-CES 118.1, Kimball baseline 105.1.	62
21	Plant-clustered bootstrap inference, 172 converged replications of 200 attempted. Each replication redraws plants with replacement, re-runs the Hsieh–Klenow cleaning including its sample-dependent trims, re-estimates the demand elasticity and the superelasticity, and re-solves the Kimball fixed point and both counterfactuals. Point estimates are the full-sample values. Intervals are percentile intervals and are conditional on convergence; Section 6.4 explains why that qualification matters.	63
22	Misallocation by state, pooled 2012–2022: within-industry variance ($\times 100$) of log TFPR, MRPL, MRPK. Ranked by var TFPR.	76
23	Within-industry variance ($\times 100$) of TFPR under gross-output and value-added accounting, same sample, same technology.	77

List of Figures

1	Within-industry dispersion of revenue productivity and of the two factor marginal products, 2012–2022. Standard deviations of log deviations from industry aggregates. Capital-wedge dispersion exceeds labour-wedge dispersion in every year by roughly half again, and all three series decline over the decade.	33
2	Within-industry variance of log TFPR, 2012–2022, computed two ways on the same plants with the same technology: value added net of materials, and gross output. The gap is the mechanical exploit of subtracting a materials bill that is 70–90 per cent of revenue. Both series fall over the decade, so the trend is invariant to the choice; the level is not.	35
3	Left: autocorrelation of each within-industry log wedge at lags one to five. Under classical measurement error these lines would lie on the dashed rule. Right: the permanent share of variance, shown as a band between the five-year covariance ratio and the between-plant variance share, because the data bound this quantity rather than identify it.	38

4	Left: the within-industry correlation between plant log TFPQ and log TFPR, and between log TFPQ and the capital wedge. Zero is the undistorted benchmark. Right: the fall in dispersion among all census-scheme plants against the fall among the 10,988 present in both 2012 and 2022. The open marker is the 2012 dispersion of plants that do not survive to 2022.	42
5	The two calibration inputs by year: the aggregate Lerner markup (left) and the demand elasticity it implies (right). The open marker is 2019, whose calibration lies far outside the other ten years and which the paper treats as a boundary case of the model class rather than as a finding (Section 6.7).	43
6	Counterfactual TFP gains from within-industry marginal-product equalisation, by reference year, each year recalibrated through ς with the superelasticity pooled. The shaded band between the two series is the markup tax on the cure: the part of the constant-markup (CES) promise that variable markups price away. Open markers are the 2019 boundary case.	45
7	Share of within-industry TFPR variance explained by seven employment-size bins, by year. The size-dependent component of wedges stays between two and six per cent throughout, with no upward trend.	52
8	Within-industry TFPR dispersion by state, pooled 2012–2022. Darker = more misallocated.	53
9	The same quantity ranked: within-industry variance of log TFPR by state, pooled 2012–2022. The dashed rule is the unweighted mean across states.	54
10	Monopsony incidence from the companion paper against this paper’s factor-wedge dispersion, 21 states. Left: the labour wedge. Right: the capital wedge. Dashed lines are ordinary least-squares fits. The two measures are built by different methods from the same plants; they agree on labour and not on capital.	57

1 Introduction

How much of India’s measured manufacturing misallocation is a real distortion, and how much of the gain from removing it could policy actually capture? We answer both questions on a new plant panel covering 2012 to 2022.

The accounting behind the first question is well established. When one plant’s marginal revenue product of capital exceeds its neighbour’s, moving capital between them raises output. No new technology is required and no new investment. An economy that leaves many such trades unexecuted produces less than it could. [Restuccia and Rogerson \(2008\)](#) and [Hsieh and Klenow \(2009\)](#) turned that observation into a measurement programme. Under explicit assumptions on technology and demand, the dispersion of revenue productivity within a narrow industry is the footprint of the distortions that drive marginal products apart. A counterfactual that equalises those products converts the dispersion into a TFP number.

For India the message was striking. [Hsieh and Klenow \(2009\)](#) put the gains from full within-industry equalisation at roughly 100 to 130 per cent of manufacturing TFP in the 1987 to 1994 cross-sections. India became the canonical high-misallocation economy, and “reallocation” a standing item on its reform agenda.

We re-measure that quantity three decades on, and we ask the question the canonical accounting cannot. What happens to the promised gains, and to their distribution, when market power is allowed to respond? The companion paper forces the question. It documents on the same plants that product-market power is modest but rises with plant size, and that labour-market power is pervasive. In the canonical framework every markup is a distortion indistinguishable from a tax, and every reallocation leaves prices unchanged. Neither assumption is innocuous here. A markup that rises with size is not a residual wedge. It is an equilibrium object, and it moves when resources move. If reallocation’s winners are the large, productive, high-markup plants, and if they re-price at their new scale, then the consequences of efficient reallocation can differ sharply from the constant-markup arithmetic. That is true of output. It is truer still of the division of income between labour and profits.

The paper asks three questions. **Q1, magnitude and anatomy:** how large is within-industry misallocation in organised Indian manufacturing over 2012–2022, how has it moved, and which factor market carries it? **Q2, the markup correction:** how much of measured dispersion is markups rather than wedges, and how much of the counterfactual gain survives when markups are modelled and allowed to respond? **Q3, incidence:** how does reallocation move the markup-implied factor shares, across the size distribution, and across space?

The identifying lever is the one this literature made standard. Under monopolistic competition and Cobb–Douglas technology, a plant’s revenue productivity is proportional

to a geometric average of its wedges. Dispersion in revenue productivity therefore measures distortion, and its factor-specific components attribute that distortion to capital or to labour. What is new is the demand side we wrap around the lever. We calibrate a Kimball system in which the markup is a known, increasing function of relative size. The markup component of measured dispersion can then be separated from the residual wedge, and the counterfactuals can let prices respond.

Four well-known problems discipline the design. **(i) Measurement error masquerades as misallocation.** Revenue productivity is a ratio of noisy survey variables, and [Bils et al. \(2021\)](#) show that much measured dispersion can be noise. Their method exploits the fact that revenue growth is less sensitive to input growth when a plant’s average products are overstated by measurement error. Applied to Indian manufacturing over 1985–2013, their correction lowers the potential gains from reallocation by **20 per cent**; for the United States the reduction is 60 per cent. Any Indian gain estimate, ours included, should be read against that figure. We trim as the literature prescribes, we privilege comparisons over levels, and we report a gross-output reconstruction that holds technology fixed and changes only the accounting. **(ii) Parameter fragility.** [Zhang and Xia \(2022\)](#) show that Hsieh–Klenow gains swing from 44 to 362 per cent across defensible parameterisations on Chinese data. We implement their programme rather than cite it. Every maintained parameter is varied in Section 6, and one instructive degeneracy is reported as a finding. **(iii) Markups contaminate the wedges.** In the canonical accounting a high-markup plant looks identical to a taxed plant. The Kimball structure resolves that observational equivalence by theory, pinning the markup down through the plant’s relative size. **(iv) The counterfactual is model-dependent.** Kimball demand is not CES, so the liberalised equilibrium cannot be computed plant by plant. It is a fixed point in each industry’s demand index.

A fifth problem we inherit rather than solve is worth naming here, because it bounds the whole framework. The Hsieh–Klenow accounting maintains Cobb–Douglas technology, and marginal products are proportional to average products only under that assumption. [Peter & Ruane \(2026\)](#) estimate long-run elasticities of substitution between eight categories of material inputs for Indian plants, identified from the heterogeneous tariff cuts of the early 1990s, and find substitutability well *above* the Cobb–Douglas benchmark of one. In their quantitative model, misallocation of intermediate inputs alone costs 29 per cent of aggregate output, three and a half times what a complementary calibration implies. We retain Cobb–Douglas for comparability with the literature we are speaking to, and Section 5.2 shows separately what the related choice between gross-output and value-added accounting is worth. Neither exercise substitutes for relaxing the technology, which we do not do.

The stakes are Indian and current. Labour’s share of value added in registered manufacturing fell by thirteen percentage points between 1980–81 and its 2007–08 trough,

before partially recovering (RBI, 2024; Basole & Narayan, 2020). On the narrower wage-bill construction, Shukla (2026) reports real manufacturing wages flat since about 2014 while output per worker doubled. Reallocation has been on the reform agenda throughout that period, and it is argued for on efficiency grounds with the distributional consequences left implicit. We price those consequences explicitly. That matters for a target India has signed: SDG 9.2 commits signatories to raise industry’s share of employment as well as of output (United Nations, 2015). An efficiency reform that lowers labour’s share is in tension with the employment half of that commitment even when it delivers the output half.

We follow the numerical implementation of Liang (2023), executing her replication code verbatim on the Indian panel, and we validate our pipeline against her published results decade for decade. From the raw ASI block files we construct 481,322 plant-years for 2012–2022 with value added, capital and wage bills. Of these, 429,534 survive the framework’s cleaning. That is about 39,000 plants a year, in every organised manufacturing industry and every state. Seven sets of findings emerge.

First, misallocation is large but improving, and capital carries it. The variance of within-industry log revenue productivity falls from 0.73 in 2012 to 0.59 in 2022. That is high by international standards throughout, but it is on a clear downward path that accelerates after 2018. The factor anatomy is lopsided. Capital-wedge dispersion (s.d. ≈ 1.1 – 1.2) is half again labour-wedge dispersion (s.d. ≈ 0.7 – 0.8) in every year, every state, and on both sides of the rural–urban divide. Counterfactual gains from full equalisation run from 121 to 184 per cent across annually recalibrated models. They sit above Liang (2023)’s 92 to 110 per cent for the 2000s and join her series smoothly at the overlap, and they trace an inverted U that peaks in 2016 and falls steeply after 2020.

Second, markups are a small share of the disease and a large tax on the cure. Modelled markups account for only about 1.6 per cent of revenue-productivity dispersion, because residual capital and output wedges dominate. Yet the Kimball market counterfactual delivers only 81 to 93 per cent of the gain a constant-markup model promises. Reallocation concentrates sales in plants whose demand is less elastic, and their re-pricing taxes away 7 to 19 per cent of the canonical arithmetic.

Third, the markup-implied labour share falls when allocation improves. Equalising marginal products lowers it by 2.65 percentage points in the Kimball economy, and by 0.94 under domestic capital elasticities (Section 6.1). Freezing markups at a constant makes the same experiment distributionally neutral, at -0.12 points. The comparison isolates the mechanism. It is the endogeneity of markups, not their heterogeneity, that turns reallocation against labour. The planner benchmark shows the converse: marginal-cost pricing raises the same share by 8.7 points before any reallocation occurs. Throughout, the wage is normalised and factor supplies are fixed, so what moves is the share of revenue the model’s markups leave to factors, not what any worker is paid; Section 5.8 is explicit

about the distinction.

Fourth, size-dependent wedges are a formality-boundary phenomenon. Transplanting the diagnostics of Hsieh et al. (2026) into the organised sector, plant size explains only 2 to 6 per cent of revenue-productivity dispersion. Neutralising the size-fitted component yields output gains of 1.2 to 1.9 per cent, under one per cent of the full-equalisation benchmark computed with the same machinery. The largest plants are, if anything, wedge-favoured. What this does *not* establish is where India's tax on size does live: the ASI observes only registered factories, so the absence of a gradient inside that range is silent about the formal-informal margin below its floor. The finding is that the size-dependent component emphasised by Hsieh et al. (2026) is thin *within registered manufacturing*; locating it elsewhere would need unorganised-sector data this paper does not use.

Fifth, the distortions fall on the productive, and half the decade's improvement is selection. Within industries the correlation between a plant's revenue-recovered physical productivity (TFPQ) and its revenue productivity is 0.696, with a slope of 0.397. That correlation is zero in an undistorted economy. It needs one qualification stated at the point of use rather than in an appendix: no plant-level price or physical quantity is observed in the ASI, so TFPQ here is not measured but recovered from revenue through the calibrated demand elasticity, which means it shares both its revenue numerator and its model parameters with TFPR. Some positive association between the two is therefore mechanical, and the figure should be read as demand-adjusted revenue productivity co-moving with revenue productivity rather than as an independent measurement of physical efficiency. What the number does establish is that the co-movement is far from the zero the undistorted benchmark predicts; what it cannot establish, without prices, is how much of the gap is distortion and how much is the shared construction. India's wedges are therefore not merely wide but aimed at the plants that would use resources best, which is why its measured gains exceed those found for China and for India's own 2000s. The correlation is also falling. Of the decade's 20.6-point fall in dispersion among census-scheme plants, 57 per cent occurred within plants observed in both 2012 and 2022, and the remainder is composition. Plants in their final observed year carry a mean absolute wedge of 0.761, against 0.601 for continuing plants, so exit falls disproportionately on the badly allocated.

Sixth, the wedges are persistent, and capital's most of all. Measured dispersion could be mismeasurement or transitory shocks rather than distortion, which would make the counterfactual gains an artefact. It is not. The one-year autocorrelation of the plant-level wedge is 0.71, and it is still 0.46 after five years. A plant in the worst-distorted fifth of its industry has a 67 per cent chance of being there again a year later, against 20 per cent under independence. The capital wedge is the most persistent on every measure. That corroborates the first finding in a way its level alone could not, because mismeasurement can inflate a level but cannot survive five years. Between two-fifths and three-quarters of

wedge variance is a fixed feature of the plant.

Seventh, labour-market power sits where the labour wedge is widest. Matching our state panel to the companion paper’s monopsony geography, the incidence of $\psi > 1$ correlates +0.54 ($p = 0.011$) with labour-wedge dispersion and +0.11 ($p = 0.62$) with capital-wedge dispersion. The agreement is not built in. One measure comes from an estimated production function and the other from differencing revenue productivities against industry aggregates, and they share no machinery. It also explains why the two papers’ headline geographies look different. The aggregate is dominated by capital, which monopsony does not touch.

We also supply inference of a kind this literature rarely reports. Counterfactual gains are constructed from parameters estimated on the same sample, so we bootstrap the whole pipeline. Each replication resamples plants and re-runs the cleaning, the demand calibration, the superelasticity regression and the Kimball fixed point. The resulting uncertainty is not uniform across our claims. Section 6.4 says which of them it leaves standing and which it does not.

A further contribution is one of measurement, and it bears on every Hsieh–Klenow study of India rather than only on ours. Holding the sample and the technology fixed and varying only the accounting, the within-industry variance of value-added revenue productivity is between 6.8 and 8.1 times its gross-output counterpart (Section 5.2). The canonical framework is value-added throughout. A large part of the headline magnitude this literature reports for India is therefore a consequence of subtracting a materials bill worth 70 to 90 per cent of revenue. The trend is invariant to the choice. The level is not. This supports on modern data the argument Kabiraj (2020) makes for gross-output accounting.

The paper contributes to three literatures. To the *misallocation measurement* literature it adds the first annually recalibrated variable-markup accounting for India. Eleven cross-sections, each with its own demand calibration, turn the single-year snapshots of the prior literature into a time path. It also implements the Zhang and Xia (2022) sensitivity programme end-to-end on Indian data, including a degeneracy in which domestic cost-share elasticities make the demand system uncalibratable, which vindicates Hsieh and Klenow’s original argument for external elasticities in an unexpected way.

To the *markups and macro* literature following Edmond et al. (2015), Peters (2020) and Edmond et al. (2023), it contributes a sharp decomposition of the two roles of markups. They are negligible in the variance accounting and decisive in the counterfactual. It adds the labour-share reversal, which extends Zhang and Xia’s China finding from a dampening to a sign flip once markups re-optimize. To the *Indian development* literature (Hsieh and Klenow, 2009; Bollard et al., 2013; García-Santana and Pijoan-Mas, 2014; Duranton et al., 2015; Hsieh et al., 2026), it updates the central estimate a decade forward on a fresh panel, relocates the size-dependent component of the problem to the formality boundary, and

documents that the much-invoked reallocation reform would, within the organised sector as it stands, be a reform against labour’s share of income.

Table 1 collects the headline numbers. Section 2 situates the paper; Section 3 describes the data; Section 4 develops the framework equation by equation; Section 5 presents results; Section 6 the sensitivity analysis; Section 7 draws the interpretation together; Section 8 concludes. Appendices document the sample (A), the numerical implementation (B), the full state panel (C), and the gross-output reconstruction (D).

Table 1: Findings at a glance (2012–2022 unless noted; “gain” = counterfactual TFP gain from within-industry marginal-product equalisation).

Object	Estimate
Estimation sample	429,534 plant-years ($\approx 39,000$ /year)
$\text{var}(\log \text{TFPR}) \times 100$, within NIC-4	72.9 (2012) $\rightarrow$ 58.6 (2022)
s.d. MRPK vs s.d. MRPL (2017)	1.15 vs 0.74 — capital wedges dominant
Markup share of TFPR variance	$\approx 1.6\%$ (baseline); 0.4% (own- α)
Kimball market gain (2017 reference)	175%; planner 179%; CES 189%
Per-year gains (recalibrated)	121–184% among plants holding half of value added; inv
Kimball/CES gain ratio	0.81–0.93 — markups tax the cure 7–19% (sign reverses unde
Δ markup-implied labour share, equalisation	–2.65pp (Kimball; –0.94 own- α) vs –0.12pp
Δ same share, marginal-cost pricing	+8.7pp (before reallocation; wage fixe
Size-fitted share of TFPR variance	2.2% (2012) $\rightarrow$ 5.6% (2022)
Size-neutralising counterfactual gain	1.2–1.9% per year
Most/least misallocated states	Odisha, AP, WB (>75) / Punjab, Haryana
$\text{corr}(\ln \text{TFPQ}, \ln \text{TFPR})$ within industry	0.696 (0 if undistorted); slope 0.397
Fall in dispersion, survivors’ share	57% of 20.6 points; rest is composition
Mean wedge : exiters / entrants / stayers	0.761 / 0.712 / 0.601
Wedge persistence, ρ at 1 yr / 5 yr	TFPR 0.71 / 0.46; capital 0.80 / 0.5
Permanent share of wedge variance	capital 47–78%; labour 41–70%
Monopsony vs labour / capital wedge	$r = +0.54$ ($p=0.011$) / $+0.11$ ($p=0.62$)

2 Related literature

Why are some countries so much poorer than others? For two decades the most productive answer has been that they are not only worse at making things but worse at deciding who makes them. A country can hold its technology fixed, move capital and labour from the plants that use them badly to the plants that use them well, and come out richer. The size of that free lunch — and whether it is a lunch at all, or an artefact of how we measure — is the question this literature exists to answer, and the question this paper joins.

2.1 Two traditions: the covariance and the wedge

Measured aggregate productivity is not the average of plant productivities. It is the average plus a term that credits an economy for having put its resources in the right

hands. [Olley & Pakes \(1996\)](#) made that arithmetic operational:

$$\Omega_t = \sum_i s_{it} \omega_{it} = \bar{\omega}_t + \sum_i (s_{it} - \bar{s}_t)(\omega_{it} - \bar{\omega}_t), \quad (1)$$

where ω_{it} is plant i 's log productivity in year t , s_{it} is its share of industry activity, $\bar{\omega}_t$ and $\bar{s}_t$ are the unweighted means of ω and s across the plants in the industry, and Ω_t is the share-weighted aggregate. The second term — the covariance between size and productivity — is what the literature calls allocative efficiency. If the productive plants are also the large ones it is positive and the economy earns a premium over its typical plant; if resources sit with the unproductive it is small or negative.

This is the first of two traditions, and its virtue is that it asks almost nothing of the researcher: no demand system, no assumption about what an efficient allocation would look like. [Bartelsman et al. \(2013\)](#) took it across countries on harmonised firm-level data and found the covariance large and positive in the United States and far smaller in transition economies, which is to say that rich countries are not merely populated by better firms but are better at feeding them. Its limitation is the flip side of its modesty. A covariance is a description, not a benchmark. It tells us that an economy allocates worse than another; it cannot say how much output is on the table, because it never writes down what the right allocation would be.

The second tradition supplies exactly that, at the price of a model. Its vocabulary comes from [Chari et al. \(2007\)](#), who showed that a wide class of frictions — financing constraints, labour regulation, informational problems — shows up in a prototype growth model as though it were a tax. The frictions themselves may be invisible; their shadow is a *wedge* between a marginal product and its price, and wedges are recoverable from data even when their causes are not. [Hopenhayn & Rogerson \(1993\)](#) had already shown in general equilibrium what such a wedge costs: a firing tax worth a year's wages lowers aggregate output by a couple of per cent and drags average productivity down with it, by keeping labour attached to plants that should have shed it. [Restuccia and Rogerson \(2008\)](#) turned the insight quantitative in a heterogeneous-plant model, and the result that made the field is theirs — idiosyncratic taxes and subsidies, if they happen to be correlated with productivity, can depress aggregate TFP by 30–50 per cent, even though every plant in the economy is behaving optimally given the prices it faces.

2.2 The Hsieh–Klenow mapping, in four steps

[Hsieh and Klenow \(2009\)](#) inverted the calibrated model into a measurement device, and the inversion is worth setting out because every caveat in this paper attaches to one of its four steps.

Step one: put the wedges in the profit function. A plant i in industry s maximises

$$\pi_{si} = (1 - \tau_{si}^Y)P_{si}Y_{si} - wL_{si} - (1 + \tau_{si}^K)RK_{si}, \quad (2)$$

where $P_{si}Y_{si}$ is revenue, w the common wage, R the common rental rate, and the two wedges do the work: τ_{si}^Y raises the cost of *all* inputs in the same proportion — a tax on scale — while τ_{si}^K raises the cost of capital relative to labour. Two wedges are the most that can be identified from two first-order conditions, which is why the framework has two and not five.

Step two: read the wedges off the input shares. With CES demand of elasticity σ and Cobb–Douglas technology of capital share α_s , the first-order conditions make each wedge a function of one observable revenue share. A plant spending unusually little on capital per rupee of revenue must be facing an unusually high τ_{si}^K ; a plant that is small relative to its productivity must be facing an unusually high τ_{si}^Y . Nothing is estimated here in the econometric sense — the wedges are solved for, plant by plant, which is both the method’s economy and its exposure.

Step three: separate the two productivities. Physical productivity (TFPQ) says how much a plant makes from given inputs; revenue productivity (TFPR) says how much it sells. In a frictionless economy the two come apart completely: a more productive plant expands until it drives its own price down, and its TFPR ends up identical to everyone else’s. TFPR therefore contains no technology at all, only wedges,

$$\text{TFPR}_{si} \propto \frac{(1 + \tau_{si}^K)^{\alpha_s}}{1 - \tau_{si}^Y}, \quad (3)$$

and this is the conceptual pivot of the whole enterprise: *dispersion in TFPR is dispersion in distortions*. In the undistorted benchmark the variance of $\ln \text{TFPR}$ within an industry is zero.

Step four: price the dispersion. If $\ln \text{TFPR}$ is jointly log-normal within an industry, the output forgone has a closed form,

$$\ln\left(\frac{Y_s^*}{Y_s}\right) \approx \frac{\sigma}{2} \text{Var}(\ln \text{TFPR}_{si}), \quad (4)$$

where Y_s^* is industry s ’s output under the efficient allocation and Y_s its actual output. One variance, scaled by one elasticity, delivers a welfare number. That is the sufficient-statistic result, and it is the reason a literature exists: it converts a micro dispersion anyone can compute into a macro counterfactual everyone wants. Hsieh and Klenow’s own estimates — 86–115 per cent for China, 100–128 per cent for India, relative to US efficiency — set the scale of the debate that followed. Section 4.1 restates (2)–(4) in the notation this paper carries through, together with the gross-output rather than value-added version (Kabiraj, 2020); the forms here are the lineage, not the implementation.

2.3 The critiques that define the frontier

Equation (3) is an identity only if the model behind it is true. Everything that moves revenue productivity and is *not* a policy distortion is nonetheless scored as one, and the past fifteen years have been an inventory of those things. Four matter for our design.

Adjustment costs. Asker et al. (2014) show that a firm facing lumpy, irreversible investment and volatile productivity will display persistent MRPK dispersion in an economy with no distortions whatever. Part of any capital-heavy dispersion is a transition path, not a tax — which bears directly on this paper, since capital wedges dominate labour wedges in every year and every state of our panel.

Measurement error. If the recorded TFPR is the true one plus noise,

$$\ln \widetilde{\text{TFPR}}_{si} = \ln \text{TFPR}_{si} + e_{si} \implies \text{Var}(\ln \widetilde{\text{TFPR}}) = \text{Var}(\ln \text{TFPR}) + \text{Var}(e_{si}), \quad (5)$$

for e_{si} independent of the truth, so *measured* dispersion is an upper bound on the real thing and (4) inherits the bias one for one. Bils et al. (2021) turn this into a correction rather than a complaint, using the covariance between revenue growth and input growth to bound the noise share; the correction shrinks the implied gains substantially but does not close the US–India gap. Bun & de Winter (2019) attack the same problem from the time dimension, separating transitory wedges from persistent ones on Dutch data — a distinction we adopt, because a wedge that reverses next year is not a policy target, and it is why Section 5.3 reports autocorrelations alongside variances.

Misspecification. Haltiwanger et al. (2018) give the sharpest version: with physical output data in hand for a set of US industries, they show that much of what the revenue-based accounting labels misallocation is demand heterogeneity and technological heterogeneity that the common- σ , common- α_s assumption has nowhere to put. Carrillo et al. (2023) attack the functional forms directly, using procurement lotteries as quasi-experimental variation to estimate marginal revenue products without imposing a demand system at all.

Parameter fragility. Zhang and Xia (2022) show on Chinese data that defensible combinations of σ and returns to scale move the headline gain by a factor of eight, and conclude for domestic estimation and systematic sensitivity reporting — the protocol Section 6 follows.

We treat these four not as reasons to abandon the accounting but as reasons to privilege its *comparative* statements. A level that is contaminated by adjustment costs and noise is still informative when differenced across factors, years, states and model variants, because the contamination is common to both sides of the comparison. That is the methodological stance of this paper, and it is why the seven results in the abstract are almost all comparisons.

2.4 Variable markups: the critique that is also a mechanism

One critique deserves separating from the rest, because this paper does not merely guard against it — it measures it. The Hsieh–Klenow derivation assumes a constant markup $\sigma/(\sigma - 1)$ common to all plants. Let markups vary and they enter (3) directly:

$$\text{TFPR}_{si} \propto \mu_{si} \times \frac{(1 + \tau_{si}^K)^{\alpha_s}}{1 - \tau_{si}^Y}, \quad (6)$$

with μ_{si} the plant’s markup of price over marginal cost, and τ_{si}^K and τ_{si}^Y the same capital and output wedges as in (2). A high-markup plant now looks distorted whether or not anything distorts it. Li & Wang (2021) make the identification consequence explicit: from TFPR alone, factor-market distortions and variable markups cannot be told apart, so any number the accounting produces is a compound of the two. Edmond et al. (2015) and Edmond et al. (2023) price markup dispersion as a misallocation in its own right, and Peters (2020) makes it an equilibrium object of entry and creative destruction rather than a nuisance parameter.

Closest to us is Liang (2023), who embeds the HK accounting in a Kimball–Klenow and Willis (2016) demand system calibrated on the Indian ASI for 1998–2008 and shows that the gains promised under CES overstate what a variable-markup economy can deliver. Methodologically this paper is her decade-forward extension — her framework, our new panel, eleven annual recalibrations — plus three additions: the labour-share counterfactual with endogenous re-pricing, the triangulation of Kimball markups against production-side (De Loecker and Warzynski, 2012) and oligopoly (Atkeson and Burstein, 2008) alternatives on the same plants, and the Hsieh et al. (2026) size diagnostics.

The table makes the honest position visible. Rows one to eleven are replication and extension: the same estimator, the same moments, a later decade, and — with a super-elasticity of 0.183 against 0.14 — somewhat steeper size-dependence of markups than the earlier window showed. That is useful but it is not, by itself, a separate contribution. The claim to one rests on the last five rows, each of which requires either a panel or a second factor market and is therefore unavailable to a single-cross-section design. Whether that is sufficient is a judgement for the reader; what we can do is state precisely where the overlap ends.

2.5 The empirical record, and the India strand

Outside India the accounting has been run almost everywhere, and the useful lesson is how differently it behaves. Gopinath et al. (2017) find capital misallocation in Spain and southern Europe *rising* through a decade of cheap capital inflows, with size-dependent financial frictions the mechanism — a reminder that wedges are not a fixed feature of poor institutions but respond to macroeconomic conditions. Brandt et al. (2013) decompose

Table 2: This paper against Liang (2023), whose framework it adopts. The two studies share an estimator and a data source and differ in window, calibration inputs and the questions asked of the solved model. Liang’s entries are as cited elsewhere in this paper; cells we cannot state from the published account are marked n.r. The rows that matter for novelty are the last four: the objects this paper adds are the labour-share counterfactual, the plant-level triangulation of markups, the within/between decomposition over a decade, and the cross-paper factor-market join — none of which is available from a single-cross-section design.

	Liang (2023)	This paper
Data	India, ASI	India, ASI (Common Factory Id panel)
Window	1998–2008	2012–2022, adjoining hers
Unit and scale	Plant cross-sections	429,534 plant-years, $\approx 39,000$ per year
Reference year shown	2004	2017, plus all eleven years
Demand system	Kimball–Klenow–Willis	Kimball–Klenow–Willis (identical)
Elasticity moment	Aggregate Lerner margin	Aggregate Lerner margin (identical)
Superelasticity moment	Within-plant margin–share slope	Within-plant margin–share slope (identical)
Superelasticity estimate	0.14 (s.e. 0.005), 2001–2008	0.183 (s.e. 0.005), pooled
Firm demand elasticities	3.17–7.12 (her Table 1)	p10 4.29, median 6.87, p90 12.13
Gains, Kimball baseline	105.1 (2004)	175.1 (2017); 121–184 across years
Gains, CES benchmark	n.r. for 2004	188.6 (2017)
Markup bite on the CES gain	≈ 20 per cent	7–19 per cent, sign demand-system dependent (Section 6.3)
Alternative demand systems	translog 86.7, Dotsey–King 109.0, oligopoly-CES 118.1	translog 171.0, Dotsey–King 201.8, oligopoly-CES 205.6
Labour-share counterfactual	not reported	–2.65pp on equalisation; +8.7pp under marginal-cost pricing
Plant-level markup triangulation	not reported	Kimball vs oligopoly-CES vs production-side, same plants
Within/between over time	single cross-sections	57 per cent within survivors; remainder composition
Factor-market join	not reported	state monopsony incidence vs labour wedge, $r = +0.54$
Full-pipeline inference	not reported	200-draw bootstrap, all draws logged

Chinese factor distortions across time, space and sectors and put the non-agricultural cost at around a fifth of aggregate TFP. The surveys by [Hopenhayn \(2014\)](#) and, with development as the frame, [Ghatak and Mookherjee \(2025\)](#) set out both what the dispersion-as-distortion mapping delivers and the conditions under which it fails. The general-equilibrium extension, in which wedges propagate through an input-output structure rather than industry by industry, is [Baqaee and Farhi \(2020\)](#); we stay inside the industry-by-industry accounting, which is the cost of comparability with the literature we are speaking to. [Restuccia and Rogerson \(2017\)](#) survey the ensuing debate, and [Bergquist et al. \(2026\)](#) give the modern general statement of when marginal-product equalisation is, and is not, the right welfare benchmark.

On India, [Bollard et al. \(2013\)](#) document the puzzle that motivates much of what follows: strong within-plant productivity growth alongside stagnant allocation. [García-Santana and Pijoan-Mas \(2014\)](#) attribute part of measured misallocation to the small-scale reservation laws, largely dismantled by the time our sample period opens; [Duranton et al. \(2015\)](#) emphasise land and infrastructure. Most recently [Hsieh et al. \(2026\)](#), on combined formal-plus-informal data for 2000–2012, argue that the *size-dependent* component of Indian wedges is what reform can realistically target, and that removing it would raise output little (1–4 per cent) while raising the skill premium and lowering welfare — misallocation as the price of equality. Our size diagnostics engage their argument on the organised sector directly, and our labour-share counterfactual delivers a complementary caution through an entirely different channel: theirs runs through skill composition, ours through markups; both warn that reallocation’s efficiency gains and workers’ gains are different objects.

2.6 Indian antecedents, and the closest one

The Indian misallocation literature has largely worked within the accounting strand ([Hsieh and Klenow, 2009](#); [Bollard et al., 2013](#)), with [Das & Pant \(2006\)](#) an early attempt to measure market imperfection in Indian manufacturing structurally rather than residually, and [Kabiraj \(2020\)](#) arguing that gross-output accounting is the right frame — a point Section 5.2 quantifies. Nearest to this paper is [Gupta \(2022\)](#). He works on the same source we do — the ASI — estimates markups by the same production approach ([De Loecker and Warzynski, 2012](#)), and covers 1998–2009, which ends two rounds before our window opens. He documents that marginal costs *and* markups rise with firm size in Indian manufacturing, and identifies the mechanism from demand shocks to poor households: larger firms make better-quality goods, sell them to wealthier and less price-elastic buyers, and charge more for both reasons.

His aggregate result is the one that speaks to ours, and it is worth stating in his units rather than paraphrasing it. Using the [Petrin & Levinsohn \(2012\)](#) aggregate productivity

decomposition and a tax–subsidy policy that redistributes toward high-markup firms, he finds reallocation gains of about **47 per cent** when markups are treated as exogenous; **15.3 per cent** at the incomplete pass-through actually observed in his data; and **33.2 per cent** in the hypothetical where every firm faces the most competitive demand environment seen in the sample. The last comparison is his headline: demand-driven markup dispersion lowers the reallocation dividend by 13.7 percentage points, or about **30 per cent** of the exogenous-markup benchmark.

The parallel to Section 5.6 is close but not exact, and the difference matters. Our Kimball market counterfactual delivers 81–93 per cent of the constant-markup (CES) gain, so variable markups price away **7–19 per cent** of what a fixed-markup model promises — under this demand system; his demand channel removes about 30 per cent of his own benchmark. These are not the same number computed twice. His counterfactual is a partial-equilibrium tax–subsidy scheme scored by an aggregate productivity decomposition, ours a general-equilibrium Kimball fixed point in which the sectoral demand index itself re-solves; his markup variation is identified from quality and consumer demand, ours from the curvature of a calibrated demand system; and the windows do not overlap. A reader should not splice the two magnitudes.

What survives the differences is the direction and the order of magnitude, and that is what makes the pair informative. Two studies of Indian manufacturing, on the same survey, using the same markup estimator, on windows a decade apart, with different counterfactuals and different sources of identification, agree that letting markups respond removes a substantial share — tens of per cent, not a rounding error — of the reallocation dividend that the canonical constant-markup accounting promises. His result also supplies the microfoundation our size gradient lacks: we observe that markups rise with size and model the pattern as an equilibrium object, whereas he explains why it arises. Read together with [Liang \(2023\)](#), whose Indian estimates run through 2008 and whose 2008 endpoint our 2012 figure adjoins closely (Section 5.6), the three papers cover Indian manufacturing from the late 1990s to 2022 and tell one story: measured misallocation is large, and market power is less a component of it than a constraint on what removing it can deliver.

2.7 Where this paper sits, and what it adds

Table 3 places the studies above along the dimensions that decide whether two misallocation numbers can be compared at all. Two of those columns do most of the work. The first is the *output measure*: almost the entire literature observes revenue and not quantity, which is precisely the opening through which demand heterogeneity and markups enter (6) — [Haltiwanger et al. \(2018\)](#) is in the table partly because it is one of the few entries that can write something other than “revenue” there. The second is *which wedges are*

recovered: a paper that identifies a capital wedge alone is answering a narrower question than one that separates capital from output wedges, and a paper that also carries a markup is answering a different question again. Where we cannot state a cell with confidence we write n.r. rather than guess.

Against that backdrop the contribution of this paper is fourfold. *First*, coverage: the accounting is run on a freshly constructed Common-Factory-Id panel of roughly 39,000 organised-sector plants a year for 2012–2022, a window that begins where the Indian evidence (Liang, 2023; Gupta, 2022; Hsieh et al., 2026) ends, so the two adjoin rather than overlap. *Second*, decomposition: because wedges are recovered factor-by-factor and year-by-year, the paper reports which factor, which year and which state the dispersion lives in, and shows the capital wedge dominating everywhere — the comparative statements that survive the critiques of Section 2.3. *Third*, markups: rather than assuming (6) away or invoking it as a caveat, the paper calibrates a Kimball demand system on the same plants and finds that variable markups account for under two per cent of measured dispersion yet price away 7–19 per cent of the gains a constant-markup model promises — a small contaminant and a large correction, which is a distinction the literature has tended to blur. It also reports, rather than suppresses, the fact that this correction changes sign under two of the four demand systems we solve (Section 6.3), so the magnitude is conditional on the Klenow–Willis specification. *Fourth*, distribution: the labour-share counterfactual asks who collects the dividend, and finds that equalising marginal products *lowers* the markup-implied labour share, so the efficiency case for reallocation and the distributional case for it point in opposite directions.

3 Data

3.1 Source and sample

The data are the plant panel constructed in working paper 1 from the raw block-level files of the Annual Survey of Industries with common factory identifiers: 481,322 plant-years over 2012–2022 with the value-added objects this framework requires. The framework uses *nominal* values — every comparison it makes is within an industry-year cell, so industry-year price levels cancel by construction; this is the standard HK justification and it spares the analysis any deflator error. Plant value added is ex-factory output less materials; capital is the closing net book value of fixed assets (Section 6.2 tests what that measurement choice costs); labour input is the wage bill (total emoluments), so that cross-plant differences in hours, skill and effort are absorbed into efficiency units. It follows, and should be said once plainly rather than left for the reader to infer, that the object this paper calls MRPL is a *labour-cost* wedge: with the wage normalised, $Py/(wL)$ is an inverse labour-cost share, and it coincides with a physical marginal revenue product

Chari et al. (2007)	US aggregate time series	n.r.	Aggregate output	Efficiency, labour, investment, government wedges	Business cycle accounting	Establishes the wedge language: diverse frictions appear in a prototype model as taxes
Restuccia and Rogerson (2008)	Calibrated heterogeneous-plant model	—	Model output	Idiosyncratic output taxes and subsidies	Quantitative model	Distortions correlated with productivity cut aggregate TFP 30–50 per cent
Bartelsman et al. (2013)	Harmonised firm data, US and Europe	1990s	Revenue	Size–productivity covariance (OP gap)	Cross-country OP decomposition	The covariance is large in the US, far smaller in transition economies: allocation and selection, not technology
<i>Panel B. The Hsieh–Klenow accounting and the critiques that define its frontier</i>						
Hsieh and Klenow (2009)	China ASIF; India ASI; US Census of Manufactures	1987–2005 (varies)	Revenue	Capital wedge τ^K , output wedge τ^Y	CES demand, Cobb–Douglas technology, TFPR dispersion	Equalising TFPR raises TFP 86–115 per cent (China) and 100–128 per cent (India)
Asker et al. (2014)	Multi-country firm panels	n.r.	Revenue	Capital wedge	Dynamic investment model with adjustment costs	Lumpy investment plus volatility generates persistent MRPK dispersion with no distortion present
Brandt et al. (2013)	China, provincial and sectoral data	1985–2007	Revenue	Capital and labour wedges across space and sectors	Wedge accounting	Non-agricultural factor distortions cost roughly a fifth of aggregate TFP
Gopinath et al. (2017)	Spain and southern Europe (ORBIS)	1999–2012	Revenue	Capital wedge (MRPK dispersion)	HK accounting plus a model of size-dependent financial frictions	Capital misallocation <i>rose</i> as cheap capital flowed in; frictions scale with size
Haltiwanger et al. (2018)	US Census of Manufactures	n.r.	Revenue <i>and</i> physical quantity	TFPR dispersion	Revenue vs. quantity comparison	Much of measured misallocation is demand and technology heterogeneity, i.e. misspecification
Bun & de Winter (2019)	Netherlands, firm-level data	n.r.	Revenue	Capital and labour wedges, transitory vs. persistent	HK accounting with a persistence decomposition	Only the persistent component is a plausible policy target
Bils et al. (2021)	US Census of Manufactures; India ASI	n.r.	Revenue	TFPR dispersion, net of noise	Measurement-error bound from revenue–input growth covariance	Correction shrinks implied gains substantially; the US–India gap narrows but survives
Zhang and Xia (2022)	Chinese industrial firms	n.r.	Revenue	TFPR dispersion	Systematic sensitivity over σ and returns to scale	Defensible parameter combinations move the headline gain by a factor of eight
Carrillo et al. (2023)	Ecuador, public procurement lotteries	n.r.	Revenue	Marginal revenue products	Non-parametric, quasi-experimental	Estimates misallocation without imposing a demand system or a production function
<i>Panel C. Variable markups inside the accounting</i>						
Edmond et al. (2015)	Calibrated, open economy	—	Model output	Markup dispersion	Oligopolistic competition	Markup dispersion is itself a misallocation; trade reduces it
Peters (2020)	Model with plant-level discipline	n.r.	Revenue	Markup dispersion	Endogenous growth with creative destruction	Markup dispersion is an equilibrium object of entry, not a nuisance
Li & Wang (2021)	Analytical, with Chinese application	n.r.	Revenue	Distortions vs. markups	Decomposition of TFPR	Factor-market distortions and variable markups are <i>not</i> separately identified from TFPR alone
Edmond et al. (2023)	US (Compustat), calibrated	n.r.	Revenue	Markup dispersion and level	Calibrated macro model	Quantifies the welfare cost of markups; the misallocation channel is the larger one
Liang (2023)	India, ASI	1998–2008	Revenue	TFPR dispersion plus Kimball markups	HK accounting inside Kimball–Klenow–Willis demand	CES gains overstate what a variable-markup economy can deliver
<i>Panel D. The Indian record</i>						
Bollard et al. (2013)	India, ASI	1980s–2000s	Revenue	TFPR dispersion	HK accounting	Strong within-plant productivity growth alongside stagnant allocation
García-Santana and Pijoan-Mas (2014)	India, ASI	n.r.	Revenue	Size-dependent distortion	HK accounting with a policy experiment	Small-scale reservation laws account for part of measured misallocation
Gupta (2022)	India, ASI	1998–2000	Revenue	Markups (PF approach)	De Loecker and Warmerdam (2019)	Markups rise with size; demand-driven markup dispersion cuts the reallocation

of labour only if the wage per efficiency unit is common across plants. That condition is exactly what employer wage-setting power violates, and this paper argues elsewhere (Section 5.11) that wage-setting power is substantial in these plants. The efficiency-units convention therefore folds part of the labour market’s own distortion into the measure of labour misallocation, which is a reason to read the labour wedge as an upper bound on physical labour misallocation and to lean on its *contrast* with the capital wedge rather than on its level. Separating employment, hours and composition would require worker-level data the ASI does not carry. — an important choice, since with headcount labour a plant employing better-paid, more productive workers would spuriously show a high marginal product of labour.

3.2 Cleaning and the estimation sample

The cleaning follows Liang (2023) line by line so that results are comparable decade to decade: drop plant-years with non-positive value added, output, capital, employment, wage bill, or materials; retain industry-year (NIC-4 $\times$ year) cells with at least fifty plants; recover the model objects; trim the one per cent tails of the log deviations of TFPR and TFPQ from industry aggregates, by year. The estimation sample is **429,534** plant-years — roughly 39,000 per year, comparable in density to Liang’s 30–40,000 — spanning every organised manufacturing division and every state. Three unavoidable deviations from her implementation are documented in Appendix A with their expected direction of effect (closing rather than average capital; energy and services not netted from value added; no profit-based filter); of these, the capital-timing and filter deviations have no signed effect on within-industry-year dispersion that we can detect, while the omission of energy and purchased services does affect it: energy intensity varies across plants inside a four-digit industry-year, so the omission does not cancel in the within-cell comparison and inflates measured dispersion by an amount we cannot sign. Appendix A states this at length; it is the sharpest measurement limitation in the paper and is not claimed to be innocuous.

3.3 Parameters taken from outside the data

The industry capital elasticities α_s are taken from the U.S. NBER–CES manufacturing database, exactly as in Hsieh and Klenow (2009) and Liang (2023), on the argument — which Section 6 demonstrates empirically rather than merely asserts — that domestic factor shares confound technology with the very distortions under study. Liang’s elasticities are coded in the NIC-2004 vintage and mapped to NIC-2008 through the official division-level concordance. The rental rate is $r = 0.10$ (five per cent real rate plus five per cent depreciation, as in HK); the wage is normalised to one (only relative marginal products matter within cells). One flag for later: the tobacco division inherits $\alpha_s = 0.95$ from the near-zero U.S. tobacco labour share — the most extreme value in the mapping

— and is retained in the baseline for comparability but examined separately in Section 6.

4 Framework

This section develops the full chain from data to counterfactual, equation by equation: the technology and wedges (4.1), the demand system that lets markups vary (4.2), the identification of its two parameters (4.3), the counterfactual equilibria (4.4), and the size-wedge diagnostics (4.5).

4.1 Technology, wedges, and revenue productivity

The economy in year t (the model is solved year by year; time subscripts are suppressed) consists of S four-digit industries; industry s contains n_s monopolistically competitive plants. Factors are supplied inelastically and are sector-specific, so all action is *within*-industry reallocation — the conservative choice, since cross-industry wedges would only add gains. Plants produce with the value-added technology

$$y_{si} = A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}, \quad (7)$$

where A_{si} is physical productivity (TFPQ), K is capital, and L is labour in efficiency units (the wage bill). Equation (7) is deliberately spare: Cobb–Douglas with a common α_s within the industry means that *any* cross-plant difference in factor proportions must be explained by prices, wedges, or market power — not technology. That is what gives the accounting its bite, and what the robustness section must therefore stress-test.

Each plant faces two idiosyncratic wedges, following HK: a capital distortion $\tau_{K,si}$ scaling the relative cost of capital, and an output distortion $\tau_{Y,si}$ scaling revenue. Here they are *residual* distortions — what remains after markups are modelled explicitly in the next subsection. The plant maximises profit subject to its demand curve, and the first-order conditions invert into recovery formulas that use only observables and parameters:

$$\tau_{K,si} = \frac{\alpha_s}{1 - \alpha_s} \cdot \frac{wL_{si}}{rK_{si}}, \quad \tau_{Y,si} = \mu_{si} \cdot \frac{wL_{si}}{(1 - \alpha_s) P_{si} y_{si}}, \quad (8)$$

with μ_{si} the plant’s markup and P_{si} the price it charges, which enters here for the first time; $\tau_{K,si}$ and $\tau_{Y,si}$ are the wedges on the plant’s capital and on its output, and r and w the common rental rate and wage an undistorted plant in the industry would face. Read (8) as diagnostics. The capital wedge is identified from factor *proportions*: under (7) an undistorted plant spends $\alpha_s/(1 - \alpha_s)$ times its wage bill on capital services, so a plant whose wage-bill-to-capital-cost ratio exceeds that benchmark behaves as if capital were expensive to it — $\tau_K > 1$ — whatever the reason (credit rationing, land, uncertainty). The output wedge is identified from the *level* of the labour share relative to the markup-

adjusted benchmark: a plant whose revenue is high relative to its factor payments, beyond what its markup explains, behaves as if subsidised. Undistorted plants have $\tau_K = \tau_Y = 1$.

Revenue productivity, TFPR, the central measured object, is price times physical productivity, and combining (7)–(8) with the demand system gives

$$\text{TFPR}_{si} = P_{si} A_{si} = \mu_{si} (r \tau_{K,si})^{\alpha_s} w^{1-\alpha_s} \tau_{Y,si} \propto \frac{P_{si} y_{si}}{K_{si}^{\alpha_s} (w L_{si})^{1-\alpha_s}}, \quad (9)$$

Here P_{si} is the price plant i in industry s charges and A_{si} its physical productivity, so TFPR is price times productivity, which is what revenue data can see. y_{si} , K_{si} and L_{si} are output, capital and the wage bill. α_s is the industry’s capital elasticity, taken from outside the data, and r and w are the common rental rate and wage that every plant in the industry would face without distortions. The three remaining terms are the objects of interest. μ_{si} is the plant’s markup, $\tau_{K,si}$ the wedge on its capital and $\tau_{Y,si}$ the wedge on its output. A plant with no market power and no wedges has $\mu = \tau_K = \tau_Y = 1$, and its TFPR collapses to the same $r^{\alpha_s} w^{1-\alpha_s}$ as everyone else’s in the industry. That is why dispersion in TFPR is informative at all.

Physical productivity is not observed in the ASI. The model recovers it by inverting revenue into quantity:

$$\text{TFPQ}_{si} \propto \frac{(P_{si} y_{si})^{\frac{\sigma}{\sigma-1}}}{K_{si}^{\alpha_s} (w L_{si})^{1-\alpha_s}}. \quad (10)$$

Here TFPQ_{si} is the plant’s *physical* productivity — output per unit of input counted in quantities rather than rupees — while $P_{si} y_{si}$ is its observed nominal revenue and K_{si} and L_{si} are capital and the wage bill, exactly as in (9). The exponent $\sigma/(\sigma - 1)$ does the inversion, where σ is the elasticity of substitution across varieties within an industry. It is the single parameter on which the separation of quantity from price rests, which is why Section 6 varies it so hard. Equation (9) is the lever of the whole literature and deserves its intuition stated plainly. In an undistorted industry, TFPR is *equalised* across plants no matter how unequal their physical productivities: a more productive plant expands, and under downward-sloping demand its price falls, until price times productivity is the same as everyone else’s — high A shows up as high output, not high TFPR. Therefore any *dispersion* in TFPR within an industry is direct evidence that something — μ , τ_K , or τ_Y in (9) — prevents marginal products from equalising. The factor-specific versions

$$\text{MRPL}_{si} \propto \frac{P_{si} y_{si}}{w L_{si}}, \quad \text{MRPK}_{si} \propto \frac{P_{si} y_{si}}{K_{si}}, \quad (11)$$

are the marginal revenue products of labour and capital, each proportional to revenue over the input’s quantity because the production function is Cobb–Douglas and the elasticities are common within an industry. That proportionality is what lets revenue data speak to marginal products at all, and it is exactly what fails if technology is not Cobb–Douglas.

The two attribute the dispersion: a plant with high MRPK relative to its industry is starved of capital; one with high MRPL is starved of labour. Equation (10) inverts revenue into quantity using the demand elasticity ($y = (Py)^{\sigma/(\sigma-1)}$ up to industry constants), which is what allows a physical-productivity measure to be recovered from revenue data at all — and is also why everything downstream depends on the demand system, the subject of the next subsection. The recovered TFPQ is therefore a demand-adjusted revenue residual, not physical output per unit of input as Foster et al. (2008) measure it where prices and quantities are observed separately; its correlation with TFPR shares revenue and the calibrated elasticity on both sides, and is partly mechanical for that reason.

Aggregation and the counterfactual follow HK’s equations 15–16. Industry TFP under the actual allocation, and under the efficient one, are

$$A_s^{\text{actual}} = \left[\sum_{i \in s} \left(A_{si} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}, \quad A_s^{\text{efficient}} = \left[\sum_{i \in s} A_{si}^{\sigma-1} \right]^{\frac{1}{\sigma-1}}, \quad (12)$$

with $\overline{\text{TFPR}}_s$ the appropriately (harmonically, VA-share) weighted industry average; manufacturing TFP aggregates across industries Cobb–Douglas with value-added weights η_s , and the **gain** is $\prod_s (A_s^{\text{eff}}/A_s^{\text{act}})^{\eta_s} - 1$. The structure of (12) explains two properties used repeatedly below. A plant contributes to the loss according to the *ratio* of its TFPR to the industry mean — so it is dispersion, not levels, that costs output. And under joint log-normality the loss takes the transparent form $\ln(A^{\text{eff}}/A^{\text{act}}) \approx \frac{\sigma}{2} \text{var}(\ln \text{TFPR})$: the cost is *convex in the demand elasticity* — more substitutable varieties mean bigger losses from the same dispersion, because the efficient allocation would concentrate production more aggressively — which is why the calibration of the demand system in Section 4.3 matters as much as the measurement of dispersion itself.

4.2 From CES to Kimball: letting the markup vary

Under CES demand every plant charges $\sigma/(\sigma-1)$: markups are a constant, they difference out of TFPR dispersion, and reallocation leaves pricing unchanged. Both implications are counterfactual — the companion paper measures a positive size–markup gradient on these very plants — so we replace CES with homothetic Kimball (1995) preferences within each sector: the sectoral aggregate Y_s is defined implicitly by

$$\frac{1}{n_s} \sum_{i \in s} \Upsilon\left(\frac{n_s y_{si}}{Y_s}\right) = 1, \quad (13)$$

where Υ is increasing and concave with $\Upsilon(1) = 1$. Concavity of Υ is the economics: the marginal contribution of a variety to the sector aggregate diminishes in its relative size, so demand curves get *steeper* for plants that are large relative to their market. We adopt

the [Klenow and Willis \(2016\)](#) specification, under which the demand elasticity facing a plant with relative quantity $x_{si} = n_s y_{si} / Y_s$ is

$$\sigma(x_{si}) = \varsigma x_{si}^{-\varepsilon/\varsigma}, \quad (14)$$

governed by two parameters with clean separate jobs: ς sets the *level* of demand elasticities (the CES/HK case is the limit $\varepsilon \rightarrow 0$), and the **superelasticity** ε/ς sets the *rate* at which a plant's elasticity falls — hence its markup rises — as its relative size grows. The plant's markup follows from the Lerner condition,

$$\mu_{si} = \frac{\sigma(x_{si})}{\sigma(x_{si}) - 1}, \quad (15)$$

where x_{si} is the plant's output relative to the industry average and $\sigma(x_{si})$ is the demand elasticity it faces at that relative size. Under CES this elasticity is a constant and the markup with it; under Kimball it falls as the plant grows, so a larger plant faces less elastic demand and charges more. The markup is therefore no longer a free parameter or a residual: it is a known increasing function of relative size. This single change does three things to the accounting. It *decontaminates the wedges*: the μ_{si} term in (8)–(9) is now measured, so τ_Y is a residual distortion net of market power rather than a mixture. It *disciplines the counterfactual*: when reallocation changes a plant's relative size, (14)–(15) say exactly how its markup responds — the incomplete pass-through channel that turns out to drive the distributional results. And it *creates an estimating equation*: (15) implies a log-linear relation between a transform of the plant's profit margin and its market share, which identifies the superelasticity from panel data (next subsection).

4.3 Identifying the two demand parameters

The demand system has two parameters per year, identified from two moments, both following [Liang \(2023\)](#) exactly.

Moment 1: the aggregate profit margin pins the elasticity level. The aggregate Lerner markup

$$\mu^{agg} = \frac{\sum_{si} P_{si} y_{si}}{\sum_{si} w L_{si} / (1 - \alpha_s)} \quad \Rightarrow \quad \bar{\sigma}^{\text{target}} = \frac{1}{1 - 1/\mu^{agg}} \quad (16)$$

is an accounting object, but μ^{agg} is not a model-free one, and it is worth being exact about what it conditions on — the calibration of the whole demand system rests on it. Its numerator is observed. Its denominator is not: it grosses the aggregate wage bill up by $1/(1 - \alpha_s)$ to impute the factor payments an undistorted economy would make, so the statistic inherits the imported capital elasticities α_s , the maintained Cobb–Douglas value-added technology, and the treatment of the wage bill as labour in efficiency units. It

also absorbs, without being able to separate them, anything that drives a wedge between the observed wage bill and the competitive labour payment — employer wage-setting power above all, which the companion evidence in Section 5.11 finds to be substantial in these very plants — and anything omitted from value added, energy and purchased services included. A high μ^{agg} is therefore consistent with product-market power, with labour-market power, or with an understated input bill, and (16) cannot adjudicate between them. That the domestic capital elasticities imply a markup below one, and so an inadmissible demand elasticity, is accordingly not evidence against the domestic shares: it shows that the calibration is being selected by the maintained α_s , which is why Section 6.1 reports the alternative rather than discarding it. Read as an accounting ratio conditional on those choices, μ^{agg} is the economy-wide ratio of revenue, $\sum_{si} P_{si} y_{si}$, to the imputed cost base built from the wage bill $\sum_{si} w L_{si}$. The object $\bar{\sigma}^{\text{target}}$ it implies is the demand elasticity a single firm charging that margin would have to face. Equating it to the margin of a representative firm facing the sales-weighted average demand elasticity gives the target $\bar{\sigma}$; the parameter ς is then solved — by an outer secant loop around the full within-industry equilibrium computation — so that the model’s sales-weighted average elasticity hits the target. The logic is worth pausing on: a decade in which manufacturing margins compress is, through (16), a decade in which calibrated demand becomes more elastic, and through the convexity of the loss in σ , a decade in which the *same* measured dispersion costs more output. The calibration table in Section 5.6 shows precisely this co-movement, and it is why we recalibrate annually instead of importing one year’s parameters into another’s dispersion.

Moment 2: the within-plant margin–share slope pins the superelasticity.

The Klenow–Willis structure implies the estimating equation

$$\frac{1}{\mu_{sit}} + \ln\left(1 - \frac{1}{\mu_{sit}}\right) = \text{const} - \frac{\varepsilon}{\varsigma} \ln \lambda_{sit} + \text{plant FE} + \text{year FE} + e_{sit}, \quad (17)$$

where μ_{sit} is the plant’s accounting Lerner markup and λ_{sit} its within-industry market share. The left side is the exact transform under which Klenow–Willis demand is log-linear in the share; plant fixed effects confine identification to *within-plant* variation — a plant whose share grows must, under (14), show a predictably fatter margin — which guards against cross-sectional confounds (better plants having both high shares and high margins for technological reasons). On 237,064 usable plant-years,

$$\widehat{\varepsilon/\varsigma} = 0.183 \quad (\text{s.e. } 0.005), \quad (18)$$

where ε/ς is the ratio of the two Kimball demand parameters of (14), and is the slope coefficient of (17). It governs how fast a plant’s demand elasticity falls as it grows, so it is the single number that sets how much markups vary with size; at $\varepsilon/\varsigma = 0$ the system

collapses to CES and markups are constant. Our estimate sits against Liang’s 0.14 (s.e. 0.005) for 2001–2008: the same order, indicating somewhat steeper size-dependence of markups in the later decade. Three independent checks corroborate it: Liang reports a production-side alternative yields 0.15 in her data; the companion paper’s production-side size gradient is positive within plants; and the calibrated markup distribution (below) matches the production-side translog distribution in level and shape. Following Liang, the superelasticity is held at its pooled value across reference years, so annual recalibration operates through ς alone.

What the two sides of (17) share, and which way it biases. The regression deserves a sharper caveat than “plant fixed effects handle it”, because the left and right sides are not measured independently. The plant’s accounting Lerner markup is $\mu_{sit} = \text{VA}_{sit} / [wL_{sit} / (1 - \alpha_s)]$ and its share is $\lambda_{sit} = \text{VA}_{sit} / \sum_j \text{VA}_{sjt}$: plant value added sits in the numerator of both. Any error in VA_{sit} therefore moves the regressand and the regressor together, and the fixed effects do nothing about it, since they sweep out levels and this is a within-plant, contemporaneous problem. Given that value added is a difference of two large numbers — the point Section 5.2 makes at length, and which puts the value-added variance at seven times the gross-output one — this error is not a remote possibility.

The direction of the resulting bias can be signed. Writing the left side as $f(\mu) = 1/\mu + \ln(1 - 1/\mu)$,

$$f'(\mu) = \frac{1}{\mu^2(\mu - 1)} > 0 \quad \text{for } \mu > 1, \quad (19)$$

so f is increasing in the markup. Since $\partial \ln \mu_{sit} / \partial \ln \text{VA}_{sit} = 1$ and $\partial \ln \lambda_{sit} / \partial \ln \text{VA}_{sit} = 1 - \lambda_{sit}$, a pure value-added error traces out the slope

$$\frac{df(\mu_{sit})}{d \ln \lambda_{sit}} = \frac{1}{\mu_{sit}(\mu_{sit} - 1)} \cdot \frac{1}{1 - \lambda_{sit}} \approx \frac{1}{\mu_{sit}(\mu_{sit} - 1)} > 0 \quad (20)$$

for the small shares that dominate the sample, where μ_{sit} is the plant’s accounting markup and λ_{sit} its within-industry share as above. At the aggregate markup of 1.289 that 2017 delivers, (20) evaluates to +2.68.

The sign is what matters. Equation (17) regresses $f(\mu)$ on $\ln \lambda$ with coefficient $-\varepsilon/\varsigma$, so a *positive* mechanical covariance biases the fitted coefficient upward, which biases $-\varepsilon/\varsigma$ upward and hence ε/ς *downward*, towards zero — towards CES and constant markups. The contamination therefore attenuates: our 0.183 is a lower bound on the true ratio under this channel, not an estimate inflated by it. That is the opposite of the direction in which a shared-component problem would ordinarily be expected to flatter a paper claiming markups vary with size, and it means the markup tax reported in Section 5.6 is, if anything, understated on this account.

What the derivation cannot supply is the magnitude, which needs the share of within-

plant variance in $\ln \lambda$ that is value-added error rather than genuine movement in share. Identifying that requires an instrument, and we report a failed attempt rather than leave the gap unexamined.

Three instruments, and why none of them works. If the problem is classical error in $\ln \lambda$, the textbook remedy is a second measurement of the same share taken from a different part of the return. The ASI offers two: the plant’s share of industry *employment* (Block E) and its share of industry *capital*, neither of which passes through value added. A third candidate isolates the denominator of the share, instrumenting with leave-one-out market size $\ln \sum_{j \neq i} VA_j$, which carries none of plant i ’s own error. Table 4 reports all three, and the two overidentified combinations, with plant and year effects absorbed as in (17).

They are not weak, and they do not agree. First-stage F statistics run from 15,673 to 50,374, so identification is not the issue. But the three estimates are -0.158 , $+0.019$ and -0.035 against an OLS $+0.183$, and the overidentification test rejects at the limit of machine precision (Hansen $p = 6.7 \times 10^{-290}$ on the two physical shares; indistinguishable from zero on all three). At least two of the three instruments are therefore invalid.

The reason is economic, not statistical, and it generalises. A plant’s employment share, capital share and value-added share differ from one another *systematically* — through factor intensity — and not merely through measurement noise. They are not three noisy readings of one latent quantity, so instrumenting one with another does not clean the regressor; it silently changes the estimand to the elasticity of the margin with respect to a different notion of size. The same objection applies to any within-return substitute we can construct, which is why we stop here rather than search for a fourth.

What the exercise does establish is that the obvious instruments are closed off, and that the honest reading of (18) is the one this section already gives: a signed bias and an unsigned magnitude. A genuine fix needs an *excluded demand shifter* — a tariff change, an export-demand shock, a downstream expansion — that moves a plant’s share without touching the mismeasurement of its value added. The companion paper on macroeconomic determinants now assembles exactly such a panel of applied output and input tariffs at the industry level, and joining it to this design is the single most valuable extension we can identify. Estimates are in `Results/superelasticity_iv.csv`, produced by `Code/R22_superelasticity_iv.do`, which also re-derives the published OLS coefficient to eight decimal places as a check on the replication.

The calibrated model at the 2017 reference year. Solving on the mid-sample cross-section, the outer loop converges in five iterations to $\hat{\zeta} = 5.34$ (so $\varepsilon = 0.98$), with the model’s sales-weighted elasticity within 10^{-3} of the 4.457 target and the implied aggregate markup 1.315 against the data’s 1.289. The equilibrium objects are economically sensible: firm demand elasticities have a 10th percentile of 4.29, median 6.87, 90th of

Table 4: Instrumenting the market share in (17). Every column absorbs plant and year effects and clusters on the plant, exactly as the baseline does. “First-stage F ” is the Kleibergen–Paap Wald statistic. The instruments are individually overwhelming and jointly rejected, which is the point: they are not noisy readings of one latent share but measures of different notions of size.

Instrument for $\ln \lambda_{sit}$	Coefficient	(s.e.)	First-stage F	Hansen p
None — OLS, as published	+0.18315	(0.00491)	—	—
Employment share	−0.15805	(0.00452)	34,696	—
Capital share	+0.01867	(0.00487)	15,673	—
Leave-one-out market size	−0.03480	(0.00441)	50,374	—
Employment + capital	−0.09808	(0.00400)	31,598	6.7×10^{-290}
All three	−0.08551	(0.00384)	33,214	$< 10^{-300}$

12.13 (Liang’s Table 1: 3.17–7.12); Kimball markups have mean 1.186, median 1.170, and a 99th percentile of 1.461 — right-skewed with a long thin upper tail, and comparable to the production-side translog markup median of 1.241 for the same plants and year. That two entirely different identification strategies — production-side cost shares there, demand-side market-share moments here — deliver overlapping markup distributions for the same plants is among the strongest internal-validity results of the project, and Section 5.7 pushes the comparison to plant level.

4.4 Four equilibria and what their differences isolate

The **liberalisation** counterfactual sets $\tau_{K,si} = \tau_{Y,si} = 1$ for all plants, equalising marginal revenue products of both factors within each industry, and re-solves the industry equilibrium. Because demand is Kimball, this is not a formula but a fixed point: each plant’s markup, price, and relative output depend on the sectoral demand index, and the index must satisfy (13). The numerical implementation is Liang’s own (secant iterations on the demand index within the calibration loop over ς , her Mata root-finders executed verbatim), with one engineering addition documented in Appendix B. Aggregate effects are Cobb–Douglas across industries with value-added weights. Four equilibria are computed:

- **Baseline:** actual wedges, Kimball pricing; the economy as measured.
- **Kimball market:** wedges removed, plants *re-optimize* markups as their relative sizes change. This is the honest liberalisation: reallocation happens, and its winners re-price along (14)–(15).
- **Kimball planner:** wedges removed *and* marginal-cost pricing imposed. The gap to the market equilibrium isolates the incomplete-pass-through channel — what variable markups cost on top of the wedges.
- **CES market:** wedges removed with markups frozen at $\sigma/(\sigma - 1)$, calibrated to the same average elasticity — the HK benchmark, what a constant-markup model

promises from the same data.

The pairwise differences are the paper’s decomposition: Kimball-market versus baseline is the attainable gain; CES versus Kimball-market is the “markup tax” on the cure; planner versus market is pass-through; and the labour-share versions of the same four equilibria (Section 5.8) turn the decomposition distributional.

4.5 Size-dependent wedges: the HKKR diagnostics

Hsieh et al. (2026) argue that the policy-relevant component of Indian wedges is the part that varies systematically with plant *size* — size-dependent regulation, enforcement, and informality — and that removing it trades efficiency against equality. Three of their diagnostics transplant onto our sample (at their $\sigma = 3$). First, the size gradient: a revenue-weighted regression of the within-industry log-TFPR deviation on their seven employment bins with NIC-4×year effects. Second, the variance decomposition

$$\text{size share}_t = \frac{\text{var}(\widehat{\text{fit}}_{it})}{\text{var}(\ln \text{TFPR}_{it})}, \quad (21)$$

the share of within-industry TFPR variance explained by the size-bin fit. Third, the **size-neutralising counterfactual**: remove only the size-fitted component of wedges (mean-preserving within industry) and recompute (12) — their Section 5.1 exercise, answering “what would fixing size-dependent policy alone be worth?” against the full-equalisation benchmark computed with the same machinery.

5 Results

5.1 The anatomy of dispersion

Table 5 reports the within-industry dispersion of log TFPR and its two factor components, year by year, on the US- α value-added pipeline.

Three facts organise the table, and Figure 1 shows them at a glance. **First, the level.** A TFPR variance of 0.59–0.73 means the plant at one standard deviation above its industry mean has revenue productivity more than twice the plant one standard deviation below — enormous unexecuted trades by any international standard, and the raw material for the large counterfactual gains below. **Second, the trend.** Dispersion falls monotonically-with-wiggles from 72.9 (2012) to 58.6 (2022) — a fifth of measured mis-allocation gone in a decade. The decline is not an artefact of sample growth (the cell structure is fixed and N rises smoothly) and survives every re-measurement in Section 6. **Third, the factor anatomy.** The dispersion of MRPK is half again that of MRPL in every single year, around 1.15 against 0.74, and the two series never come close to crossing

Table 5: Within-industry (NIC-4 $\times$ year) dispersion of revenue productivity and factor marginal products, by year. Standard deviations of log deviations from industry aggregates; final column is $\text{var}(\log \text{TFPR}) \times 100$.

Year	N	s.d. MRPL	s.d. MRPK	s.d. TFPR	$\text{var} \times 100$
2012	34,595	0.806	1.195	0.854	72.9
2013	36,152	0.783	1.199	0.849	72.0
2014	37,942	0.763	1.191	0.830	68.9
2015	37,550	0.760	1.177	0.826	68.3
2016	39,776	0.742	1.159	0.805	64.7
2017	39,305	0.736	1.146	0.798	63.7
2018	39,426	0.709	1.113	0.766	58.6
2019	39,817	0.721	1.120	0.775	60.0
2020	40,830	0.722	1.112	0.776	60.2
2021	41,519	0.713	1.101	0.764	58.4
2022	42,622	0.710	1.107	0.766	58.6

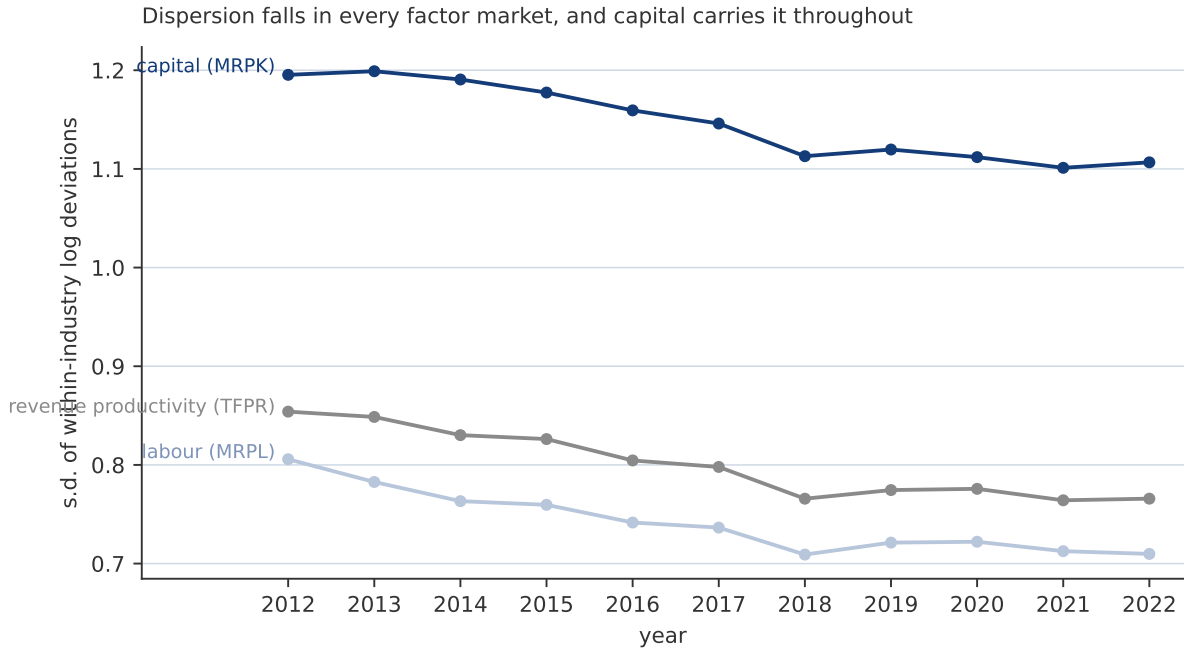


Figure 1: Within-industry dispersion of revenue productivity and of the two factor marginal products, 2012–2022. Standard deviations of log deviations from industry aggregates. Capital-wedge dispersion exceeds labour-wedge dispersion in every year by roughly half again, and all three series decline over the decade.

(Figure 1) — so the Indian allocation problem is first a *capital*-market problem: credit, collateral, land, and vintage frictions that drive plants’ shadow costs of capital apart. The labour margin is tighter but far from clean, and the companion paper’s finding of monopoly in a large minority of plants gives part of that residual a name and a mechanism. Separating a genuine factor-market distortion from a variable markup is the identification problem Li & Wang (2021) formalise, and the Kimball structure is how we address it: the markup is modelled rather than left in the residual, so its contribution can be read off instead of assumed. Within this dispersion, the modelled Kimball markups account for only ≈ 1.6 per cent of TFPR variance (0.4 per cent under domestic elasticities): consistent with the companion paper’s finding that markups are modest and their dispersion tight, market power is a *small share of the disease*, though, as Section 5.6 shows, a large tax on the cure.

That share should be read as a lower bound, for a reason internal to the model. The Kimball structure makes the markup a function of relative size, so the only markup variation it can attribute to markups is variation that tracks size. Holguín et al. (2026) show this is the smaller part. Decomposing the variance of log markups within and between bins of market share, they find that over half of markup dispersion is between firms of *similar* size: 57 per cent in Indian revenue data, and over three-quarters for price-based markups in Indian product markets computed from product-level prices and quantities. Models in which markups depend only on size, whether through concentration as in Atkeson and Burstein (2008) or through demand curvature as in Kimball (1995) and here, capture less than half of observed markup variation and assign most of what they do capture to differences between size classes. When the within-size component is modelled instead, the implied output loss from markup dispersion rises five to ten fold. We cannot recover that component with the demand system we calibrate, and we do not claim to. What we can say is that 1.6 per cent is what markups explain *once markups are required to be a function of size*, and that the true share is larger.

5.2 How much of the measured problem is the accounting?

Before any counterfactual is run, one measurement question has to be settled, because it governs the size of everything that follows. The canonical framework is a *value-added* framework, and value added is a difference of two large numbers: subtracting materials that run to 70–90 per cent of revenue for most plants mechanically amplifies whatever noise and whatever true dispersion the underlying data contain. If that amplification is large, then the headline magnitudes of the entire Hsieh–Klenow literature on India — all of it value-added — are partly an artefact of an accounting choice rather than a fact about Indian factor markets.

The question is answerable on our own data, and cleanly, because we can hold ev-

everything else fixed. Appendix D reports a reconstruction in which *both* TFPR measures are built on the same sample from the same technology (the companion paper’s translog gross-output elasticities, industry means, CRS-normalised), so that the only thing that differs between them is the accounting:

$$\ln \text{TFPR}_{it}^g = \ln Y - s_K \ln K - s_L \ln L - s_M \ln M, \quad \ln \text{TFPR}_{it}^{va} = \ln(Y - M) - a_K \ln K - a_L \ln L, \quad (22)$$

with $a_K = s_K / (s_K + s_L)$.

The answer is that the accounting choice is worth a factor of seven. Figure 2 plots the two variances year by year. The value-added measure runs between **6.8 and 8.1 times** the gross-output measure, and between seven and eight times in eight of the eleven years (2017: 122.8 against 16.6, $\times 100$). This is not a subtle sensitivity that moves a third decimal. A literature that reports Indian within-industry TFPR variance near 1.2 in logs and one that reports it near 0.4 are describing the same plants.

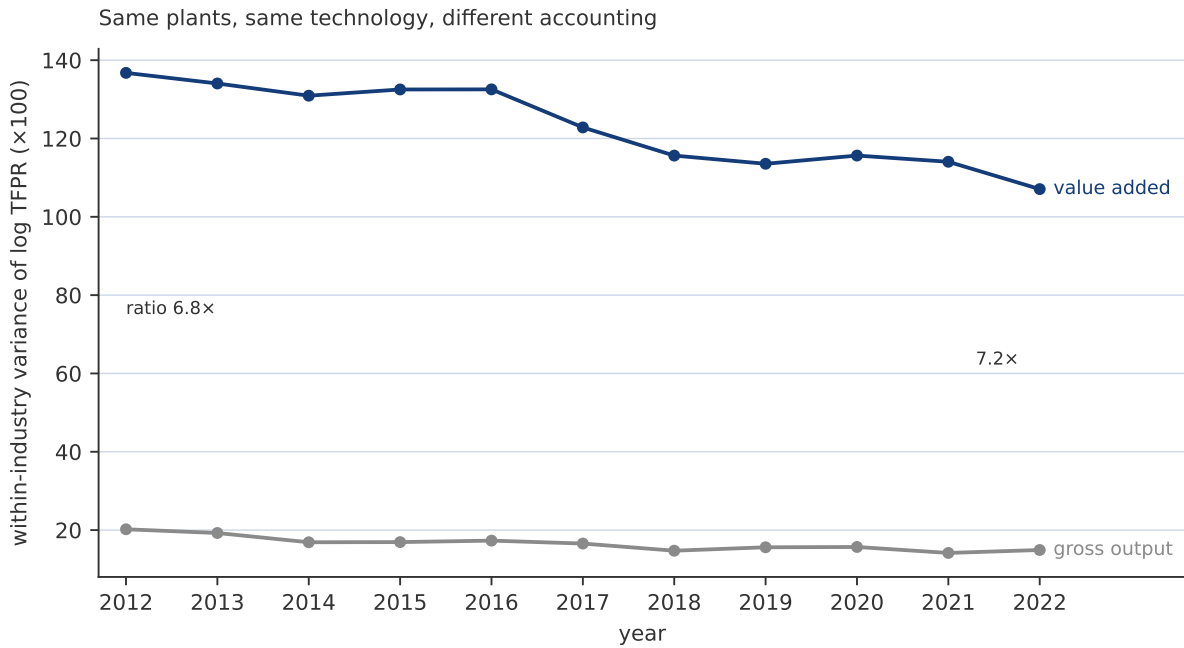


Figure 2: Within-industry variance of log TFPR, 2012–2022, computed two ways on the same plants with the same technology: value added net of materials, and gross output. The gap is the mechanical exploit of subtracting a materials bill that is 70–90 per cent of revenue. Both series fall over the decade, so the trend is invariant to the choice; the level is not.

Three consequences, in ascending order of importance. First, for this paper: the level of any value-added “gain” inherits the amplification, which is one more reason our claims ride on comparisons. That strategy has support beyond convenience. De Ridder et al. (2024) show analytically, and validate in simulation and on administrative production and pricing data, that trends in markups and their dispersion across firms are well measured from revenue or financial data, while recovering the average *level* requires price data

this kind of survey does not carry. The quantities we lean on are the ones the data can bear — across years, across states, across counterfactuals — rather than on levels. Second, the two series decline over the decade in parallel, so the paper’s central *trend* conclusion is accounting-invariant, which is exactly the sort of thing that ought to be checked rather than assumed. Third, and beyond this paper: the finding supports [Kabiraj \(2020\)](#), who argues on Indian data that gross-output accounting is the appropriate frame for misallocation measurement, and it quantifies what is at stake in ignoring the point. The same appendix confirms that the elasticity source is second-order once the accounting is fixed: our-technology value-added dispersion (s.d. ≈ 1.1) sits close to the canonical US- α pipeline value (≈ 0.80), so it is the materials subtraction and not the technology that does the work.

Why the paper nonetheless reports value added. A reader who has got this far is entitled to ask why, given a factor of seven, the headline specification is not simply gross output. The answer is comparability, and it is a choice rather than an oversight. Every Indian estimate this paper is measured against — [Hsieh and Klenow \(2009\)](#) for the founding numbers, [Liang \(2023\)](#) for the variable-markup extension whose framework we adopt and whose decade we adjoin — is value-added, and a gross-output headline would leave the paper’s central comparisons with nothing to compare against. The counterfactual machinery is also built on the value-added technology of (7); re-founding it on gross output is not a relabelling but a different model, with its own elasticity estimates and its own calibration.

The cost of the choice is stated rather than hidden: the *level* of every dispersion statistic and every gain in this paper is inflated by the materials subtraction, by a factor this section puts between 6.8 and 8.1. A reader who prefers gross output should therefore treat the levels here as literature-comparable rather than structural, and should read the paper’s conclusions — which are almost all comparisons across factors, years, states and demand systems — as the objects that survive the accounting choice, because Figure 2 shows the two series moving together. We regard [Kabiraj \(2020\)](#) as right that gross output is the better frame, and the honest position is that this paper reports the value-added numbers because they are the ones the literature can read, while documenting exactly what they cost.

5.3 How much of the measured dispersion is transitory noise?

The accounting question settled, a harder one remains, and the paper has so far deferred it to a caveat. Measured dispersion in revenue productivity need not be distortion at all. It could be measurement error in the underlying returns, or a transitory shock that the plant has not yet adjusted to, or the shadow of adjustment costs that make instantaneous

equalisation privately unprofitable rather than socially unattainable (Asker et al., 2014; Bils et al., 2021). Haltiwanger et al. (2018) put the worry at its sharpest: a residual that absorbs every unmodelled influence will always look like a distortion, so a dispersion measure earns its interpretation only when something independent of its own level corroborates it. On any of those readings the counterfactual gains of Section 5.6 are overstatements of what any policy could deliver.

The panel can discriminate, because the three readings make opposite predictions about *time*. A wedge that reflects a genuine and durable distortion — a plant that cannot borrow against the collateral it has, cannot get reliable power, cannot obtain land at the price its neighbour pays — should follow the plant from year to year. A wedge that reflects a transitory shock should not, because a transitory shock dies out, and neither should classical measurement error, which is serially uncorrelated by construction.

That second claim needs narrowing, and we narrow it rather than lean on it. Bils et al. (2021), whose challenge this subsection answers, do not mean only classical error. Their concept includes revenue or inputs omitted or double-counted across divisions or products within a firm, for instance in relation to a headquarters. A reporting bias of that kind is a fixed feature of the plant and would be *persistent*. Our test therefore discriminates against transitory mismeasurement and transitory shocks, which is what most of the literature’s noise concern is about, and it does not discriminate against a systematic, plant-specific reporting bias. We say which of the two we can exclude. So we ask how persistent the wedges are, and we ask it separately for capital and labour, which is the version that bears on this paper’s central claim. Bun & de Winter (2019) make precisely this move on Dutch data, treating the trend and the persistence of capital and labour misallocation as the two things a dispersion measure ought to report alongside its level; we follow their framing on Indian plants.

Write the wedge for plant i in year t as its log deviation from the industry-year aggregate, $w_{it} = \ln(x_{it}/\bar{x}_{jt})$, computed for revenue productivity and for each factor’s marginal revenue product. Three statistics follow, set out in Table 6. They are reported together because each is vulnerable to something the others are not: an autocorrelation is sensitive to outliers, a quintile transition throws away the size of the wedge, and a variance share depends on how many years the plant is observed for.

The answer is not close, and Figure 3 shows it. The one-year autocorrelation of the TFPR wedge is **0.71** and it is still **0.46** five years later; a plant in the most distorted fifth of its industry has a **67 per cent** chance of being there again next year and a **48 per cent** chance five years on, against 20 per cent under independence. Whatever drives Indian plants’ marginal products apart, it is overwhelmingly attached to the plant and it lasts. Measured dispersion in this panel is not mostly noise, and the counterfactuals are not mostly pricing a statistical artefact.

Two refinements matter more than the headline. First, **the capital wedge is the**

Table 6: Persistence of the within-industry wedges, 2012–2022. 429,534 plant-years on 132,196 plants; 255,385 one-year pairs and 107,713 five-year pairs. “Stay” is the probability that a plant in the top (or bottom) quintile of its industry-year wedge distribution is in the same quintile k years later; 0.20 is the value under independence. The permanent-share band runs from the five-year covariance ratio (a lower bound, since transitory shocks and classical error do not survive a five-year lag) to the between-plant variance share (an upper bound, since a plant mean over few years still carries noise).

Wedge	Autocorrelation		Stay in quintile		Permanent share	
	lag 1	lag 5	+1yr	+5yr	lower	upper
Capital, lnMRPK	0.796	0.554	0.730	0.548	47.1%	77.7%
Revenue prod., lnTFPR	0.708	0.456	0.669	0.484	40.1%	70.9%
Labour, lnMRPL	0.689	0.464	0.643	0.436	41.4%	69.9%
Under measurement error	0	0	0.20	0.20	0%	0%

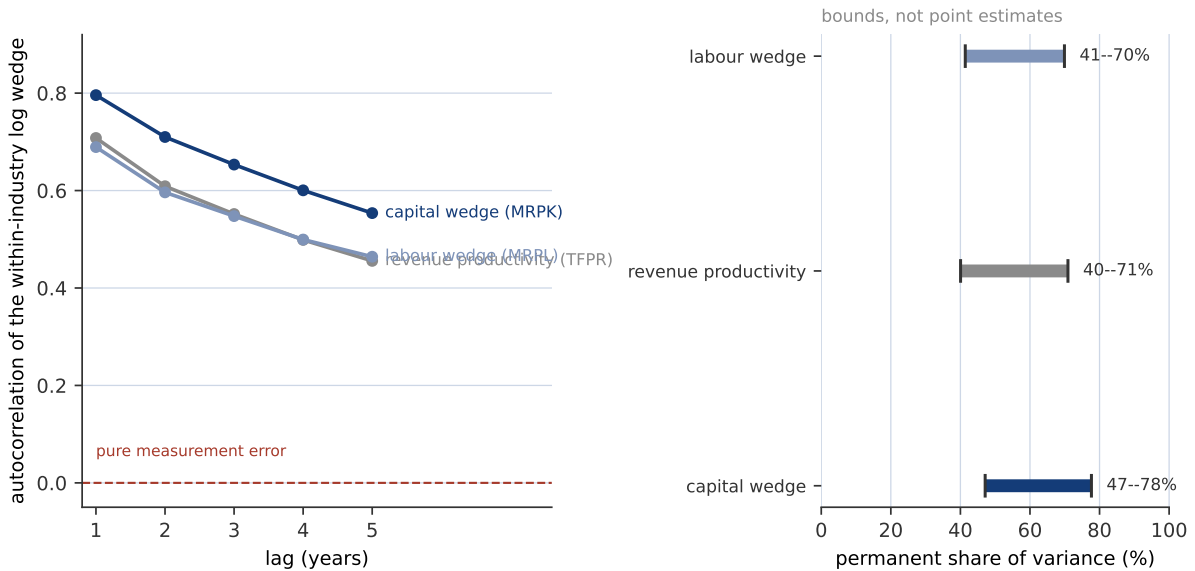


Figure 3: Left: autocorrelation of each within-industry log wedge at lags one to five. Under classical measurement error these lines would lie on the dashed rule. Right: the permanent share of variance, shown as a band between the five-year covariance ratio and the between-plant variance share, because the data bound this quantity rather than identify it.

most persistent object in the table, on every measure and at every horizon: autocorrelation 0.80 at one year against 0.69 for labour, a 73 per cent chance of staying in the worst quintile against 64 per cent, and a permanent share bounded between 47 and 78 per cent against 41 and 70. This is independent corroboration of Section 5.1’s headline, and it is corroboration of a different kind: that section established that capital wedges are *larger*, this one that they are *more durable*. Size alone is consistent with capital being measured more noisily than labour, which would have been an uncomfortable alternative reading, since capital is the harder of the two to measure. Durability is not: mismeasurement does not persist for five years.

The credit, collateral and land frictions that the Indian literature has long emphasised (Banerjee & Duflo, 2005; Das & Pant, 2006) are the kind of thing that does, and so is unreliable power. Allcott et al. (2016) instrument Indian electricity shortages with hydroelectric supply shifts and find that average reported shortages cost the average plant five to ten per cent of revenue and producer surplus. Measured productivity losses are smaller, because most inputs can be stored through an outage, but shortages distort the plant size distribution. That is a friction attached to a plant’s location and durable in exactly the way our persistence statistics describe.

The strongest evidence that these frictions are the binding ones is causal rather than descriptive. Bau & Matray (2022) exploit India’s staggered automatic approval of foreign direct investment, up to 51 per cent of domestic equity, across disaggregated industries over the 2000s. Domestic firms that began with high marginal revenue products of capital respond by raising revenue 23 per cent, physical capital 53 per cent and wage bills 28 per cent relative to low-MRPK firms, and their MRPK falls by 33 per cent. The effects are largest where the local banking sector is least developed, which points at the friction directly. Treated industries’ Solow residual rises by 3 to 16 per cent. Their design identifies what our panel can only measure, and the two fit: a wedge that a credit-market reform compresses is a wedge that persists while credit markets do not change. Credit and collateral frictions are what the quantitative literature on capital misallocation identifies as its principal sources: David and Venkateswaran (2019) attribute the bulk of measured capital-wedge dispersion to persistent firm-specific distortions rather than to adjustment costs or informational frictions, and Buera et al. (2011) and Midrigan and Xu (2014) show in calibrated models how financial frictions generate exactly this pattern — large, durable gaps in the marginal product of capital that entrants and small firms cannot arbitrage away.

Second, **the labour wedge is the least persistent**, and that fits rather than embarrasses the rest of the paper. Section 5.11 showed that labour-wedge dispersion is where the companion paper’s monopsony lives. Employer wage-setting power in a thin local labour market is a frictional phenomenon that responds to entry, to a new road, to an outside offer; it should be less permanent than a collateral constraint, and it is. The two

findings are the same fact seen from two directions.

Finally, the bounds are bounds and we report them as such. The lower figure is a five-year covariance ratio, which is conservative because a genuinely permanent component would show up fully at any lag while anything transitory has already dropped out; the upper figure is the between-plant variance share, which is generous because a plant mean taken over a handful of years still carries some noise. The honest statement is that between two-fifths and three-quarters of measured wedge variance is a fixed feature of the plant, and that the share is highest for capital. One further qualification is owed, and it narrows what persistence proves. [Bergquist et al. \(2026\)](#) distinguish *technological* wedges, which are present even in the allocation a social planner would choose, from *distortionary* ones, which appear only in market equilibrium, and they insist that not every wedge is distortionary. Persistence discriminates sharply against measurement error and against transitory shocks, because neither survives a five-year lag. It does not discriminate against a technological wedge. A plant that is durably different in its production technology, its product mix or its input quality would show exactly the persistence we document and would not represent a misallocation at all. What the result establishes is that measured dispersion is a durable feature of plants rather than noise. Which durable feature it is remains open. Nothing in this subsection identifies *which* friction is responsible, and we do not claim it does.

5.4 Who bears the wedges, and where the improvement came from

Two questions remain about the dispersion this section has measured. Neither is answered by its level or its persistence, and both change how the counterfactual should be read.

The distortions fall on the productive. The cost of misallocation depends on more than how far apart marginal products are. It depends on *which* plants sit at the wrong end. A distortion system that scatters at random wastes resources; one that systematically burdens the plants that would use resources best wastes far more, which is why [Restuccia and Rogerson \(2008\)](#) built their model on idiosyncratic taxes *correlated with productivity* rather than merely dispersed. The diagnostic is direct. In an industry without wedges, revenue productivity is equalised across plants, so within an industry the correlation between physical productivity and revenue productivity is zero by construction. Any positive value is the distortion system tracking productivity.

In Indian manufacturing that correlation is **0.696** pooled across 2012–2022, and the regression of the log TFPR deviation on the log TFPQ deviation has a slope of **0.397**: a plant one log point more productive than its industry carries about four-tenths of a log point more revenue productivity, which under the accounting is four-tenths of a log point

more wedge. The relationship holds against the capital wedge on its own at 0.57. This is a large number, and it is the part of the Indian result that the dispersion statistics alone do not convey. India’s measured gains exceed those Liang finds for the 2000s and those [Hsieh and Klenow \(2009\)](#) find for China not only because the wedges are wide but because they are aimed.

It is also falling. The correlation runs from **0.724** in 2012 to **0.685** in 2022 (Figure 4, left), and its decline is part of why the counterfactual gains decline: the system is becoming not only less dispersed but slightly less well aimed at the productive. We report this as an accounting relationship, not a causal one. Nothing here says that being productive causes a plant to be taxed; a common third factor — size, age, visibility to the tax authority, dependence on external finance — could generate the same correlation, and distinguishing those is beyond this design.

Where the improvement came from. Measured dispersion falls by a fifth over the decade and Section 5.1 called that the sample’s good news without asking what produced it. Two mechanisms would look identical in the aggregate: surviving plants could have become better allocated, or the badly allocated ones could have disappeared. The first is a system improving, the second is selection doing the work while the survivors stand still.

Separating them requires care with this survey, and the care is the reason for the restriction that follows. A plant’s absence from an ASI round does not by itself mean it stopped producing: units above the census threshold are enumerated every year, while smaller units are sampled and rotate in and out of the frame. Treating rotation as exit would manufacture a selection effect out of survey design. We therefore run this exercise on census-scheme plants only, identified by a survey multiplier at or near one — 71.7 per cent of plant-years, and the larger plants, with a median of 125 persons engaged against 19 among the sampled. The finding is theirs and we do not generalise it to the rotating tail.

Among census plants, the variance of the log TFPR deviation falls from **82.4** to **61.8** ($\times 100$), a drop of 20.6 points. Among the **10,988** plants observed in both years it falls from 71.0 to 59.2, a drop of 11.8 points — **57 per cent** of the total. So a little over half of India’s decade of improvement happened *inside* surviving plants, and a little under half is composition. The composition part is not subtle: the plants present in 2012 that do not survive to 2022 had a dispersion of **90.6**, against 71.0 for those that do.

The plant-level pattern says the same thing. Ranking census plants by the absolute size of their wedge, the mean is **0.761** for plants in their final observed year, **0.712** for plants in their first, and **0.601** for continuing plants. Exit is falling disproportionately on the badly allocated, entrants arrive closer to the middle than the leavers were, and the plants that persist are the best allocated of the three groups. That is what a functioning selection margin looks like. It also has a plausible external driver: [Bau & Matray \(2022\)](#) show that

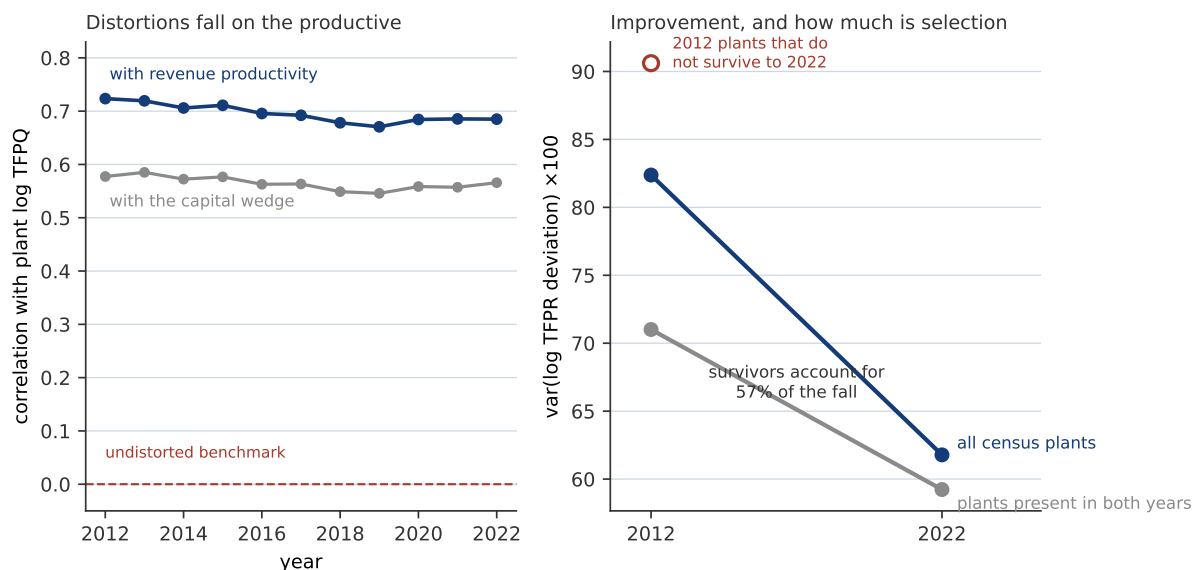


Figure 4: Left: the within-industry correlation between plant log TFPQ and log TFPR, and between log TFPQ and the capital wedge. Zero is the undistorted benchmark. Right: the fall in dispersion among all census-scheme plants against the fall among the 10,988 present in both 2012 and 2022. The open marker is the 2012 dispersion of plants that do not survive to 2022.

the capital-market liberalisation of the 2000s reduced capital misallocation in treated Indian industries, and our window opens as that process matures. We cannot attribute our decline to their policy, since our design has no treatment and no control, but the direction and the factor market agree. This is a more encouraging reading of the decade than the headline alone supports — with the caveat, which Section 5.3 makes unavoidable, that the wedges facing the plants that *do* survive are highly persistent. India’s organised manufacturing is clearing out its worst-allocated plants while the distortions facing the rest stay put.

5.5 Calibration by year

Table 7 reports the two calibration inputs for each annual model: the aggregate Lerner markup (16) and the implied elasticity target.

The margin compression of the mid-decade is clearly visible in Figure 5: the aggregate Lerner markup falls from 1.56 (2012) to 1.28–1.29 (2016–2018) before partially recovering — and maps through (16) into elasticity targets rising from 2.8 toward 4.6. The year 2019 is flagged now and throughout: its Lerner markup of 1.19 implies a target of 6.27, the edge of the model class’s solvable domain (Section 6); its calibration ($\hat{\varsigma} = 7.4$) and gains are reported but read as a boundary case, not evidence.

These aggregate markups are not the markups of the companion paper, and a reader who sets the two side by side should know why they differ in both level and trend. The companion paper estimates plant markups from a gross-output production function,

Table 7: Annual calibration targets: aggregate Lerner markup and implied sales-weighted demand-elasticity target. Comparison: [Liang \(2023\)](#), 2004: markup ≈ 1.52 , target 2.92.

Year	N (cleaned)	Aggregate Lerner markup	Target elasticity $\bar{\sigma}$
2012	34,595	1.565	2.77
2013	36,152	1.430	3.33
2014	37,942	1.370	3.70
2015	37,550	1.389	3.57
2016	39,776	1.289	4.46
2017	39,305	1.289	4.46
2018	39,426	1.279	4.58
2019	39,817	1.190	6.27
2020	40,830	1.272	4.68
2021	41,519	1.378	3.64
2022	42,622	1.356	3.81

θ^M/α^M , and its sales-weighted aggregate runs between 1.12 and 1.17. The statistic μ^{agg} in (16) is a value-added margin: since $wL/(1-\alpha_s)$ in its denominator is the wage bill grossed up to all value-added factor payments, μ^{agg} is, up to the weighting across industries, $(1-\alpha_s)$ divided by labour’s share of value added. Two things follow. Its level is higher, because a given price–cost margin is a larger fraction of value added than of gross output. And its trend is the mirror image of the labour share of value added, which rose over the decade, so μ^{agg} falls mechanically as that share rises, whereas the gross-output markup of the companion paper is close to flat. Neither contradicts the other; they measure the margin on different bases, and only the second is identified from production elasticities.

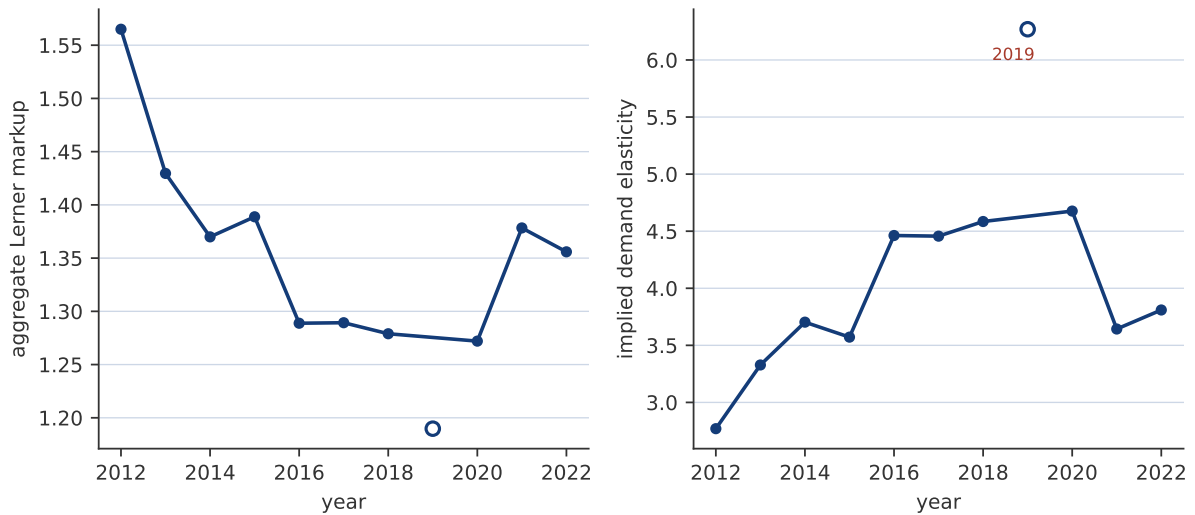


Figure 5: The two calibration inputs by year: the aggregate Lerner markup (left) and the demand elasticity it implies (right). The open marker is 2019, whose calibration lies far outside the other ten years and which the paper treats as a boundary case of the model class rather than as a finding (Section 6.7).

5.6 Counterfactual gains: level, path, and the markup tax

Table 8 is the paper’s central table: for each reference year, the annually recalibrated model’s TFP gains from within-industry marginal-product equalisation, under the three counterfactual equilibria of Section 4.4.

Table 8: Counterfactual TFP gains by reference year (per cent), each year recalibrated. Recalibration operates through the demand-elasticity parameter ς ; the superelasticity ratio ε/ς is pooled across years, following Liang (2023), so the year-to-year movement is not a full re-estimation of the demand system. “Market” lets markups re-optimize; “Planner” imposes marginal-cost pricing; “CES” freezes markups (HK benchmark). Final column: attainable share of the CES promise. 2019 is a boundary case (see text).

	Ref. year	$\hat{\varsigma}$	N	Kimball market	Kimball planner	CES	Market/CES
2017 (excl. tobacco)	2012	3.45	32,324	120.8	128.7	148.4	0.81
	2013	4.09	33,540	140.8	146.2	163.8	0.86
	2014	4.53	35,595	158.5	163.2	186.8	0.85
	2015	4.34	35,226	153.3	158.2	166.9	0.92
	2016	5.36	36,601	184.2	188.1	201.4	0.91
	2017	5.34	37,699	175.1	178.7	188.6	0.93
		4.94	37,303	158.3	162.1	169.8	0.93
	2018	5.47	38,293	161.6	165.2	181.5	0.89
	2019 [†]	7.40	38,476	270.0	274.1	311.2	0.87
	2020	5.58	38,465	170.1	173.7	196.0	0.87
	2021	4.37	40,312	126.3	130.3	142.8	0.88
	2022	4.58	40,746	128.9	132.2	140.0	0.92

What the gains cover. The N column of Table 8 is smaller than the cleaned panel of Table 7, and the difference matters more than its size suggests. The Kimball calibration of Liang (2023), which we use without modification, solves each plant’s relative output by root-finding and drops every plant for which no root exists. Table 9 shows what that removes: 3 to 8 per cent of plants in each year, but the largest ones (median employment between 120 and 845, against 44 to 86 among the plants kept), so that the dropped plants hold 21 to 26 per cent of employment and 39 to 51 per cent of value added. Every Kimball, planner and CES gain in this paper, and the labour-share counterfactual, is therefore computed on the plants that hold roughly half of organised-sector value added, with the largest plants of each industry outside it. The dispersion statistics of Section 5.1 and the persistence and geography results do not depend on the solver and cover the full cleaned panel. We do not know the direction in which the omission biases the gains: the largest plants are not unusually distorted (their mean absolute TFPR deviation is slightly below that of the plants kept in 2017), but they carry so much of value added that their absence changes the weights in every industry aggregate. The numbers should be read as gains among the plants the demand system can price, which is a narrower object than the economy-wide gains the literature, ours included, has usually reported.

Four observations organise the table, and Figure 6 plots it; all of them refer to the

Table 9: Sample flow from the cleaned panel to the counterfactual: plants the Kimball calibration cannot price. The solver of Liang (2023), used verbatim, drops any plant whose relative-output equation has no root in its search interval. Dropped shares of plants, employment and value added are of the cleaned panel for that year. The surviving N reproduces Table 8 exactly in every year. Source: `solver_dropcount.csv`, `39_solver_dropcount.do`.

Year	Cleaned	Priced	Dropped	% plants	% employment	% value added
2012	34,595	32,324	2,271	6.6	26.1	50.9
2013	36,152	33,540	2,612	7.2	26.4	50.7
2014	37,942	35,595	2,347	6.2	24.8	47.6
2015	37,550	35,226	2,324	6.2	25.6	47.8
2016	39,776	36,601	3,175	8.0	25.4	47.9
2017	39,305	37,699	1,606	4.1	22.2	44.1
2018	39,426	38,293	1,133	2.9	21.6	41.6
2019	39,817	38,476	1,341	3.4	22.0	39.3
2020	40,830	38,465	2,365	5.8	23.3	43.0
2021	41,519	40,312	1,207	2.9	20.8	44.7
2022	42,622	40,746	1,876	4.4	22.4	41.8

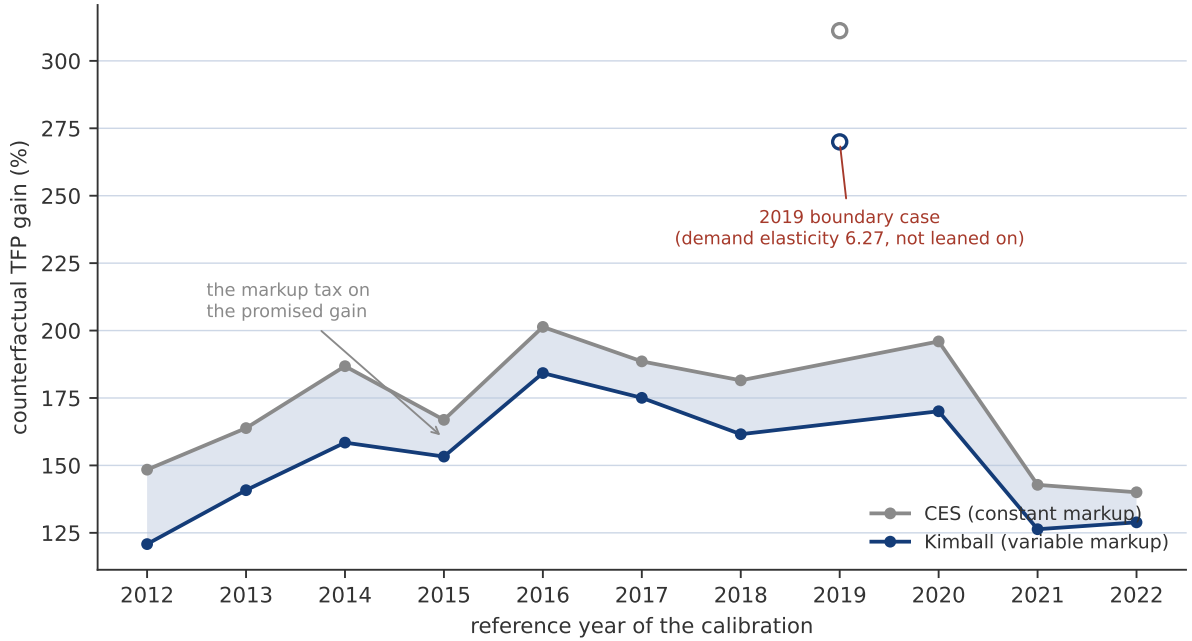


Figure 6: Counterfactual TFP gains from within-industry marginal-product equalisation, by reference year, each year recalibrated through ς with the superelasticity pooled. The shaded band between the two series is the markup tax on the cure: the part of the constant-markup (CES) promise that variable markups price away. Open markers are the 2019 boundary case.

plants the solver can price (Table 9). **First, the level.** Gains of 121–184 per cent exceed Liang (2023)’s 92–110 per cent for the 2000s, in one-to-one correspondence with the larger measured dispersion. The 2012 estimate ($\hat{\varsigma} = 3.45$, gain 121%) joins her 2008 endpoint ($\hat{\varsigma} = 3.64$, gain 92%) smoothly, which is the cleanest external validation of the pipeline available. As always in this literature the level is an upper bound twice over, since it prices complete elimination of within-industry wedges and part of measured dispersion is noise and adjustment (Section 6), so the robust content lies in comparisons.

Second, the path. An inverted U, plain in Figure 6: rising steeply to a 2016 peak (184%), declining gently to 2018, sharply after 2020, so 2021–22 gains sit near the 2012 level. Note that this is *not* the shape of raw dispersion, which falls throughout (Figure 1); the hump is contributed by the annual recalibration, which is the fourth observation below. Measured allocative efficiency worsened through the mid-decade — the demonetisation and GST-introduction window — and improved markedly through the COVID recovery, when exit and consolidation were substantial. The accounting cannot adjudicate causes, but the fact itself is new: it is invisible to the single-cross-section design of the prior literature, and it cautions against reading any one year’s “gain” as a structural constant.

Third, the markup tax on the cure. That reallocation’s welfare consequences run partly through markup *adjustment* rather than through reallocation alone is the general point Weinberger (2020) makes, deriving a sufficient statistic for allocative efficiency under variable markups and identifying it from exchange-rate shocks; here the same channel is measured directly against its constant-markup counterfactual. The Kimball market counterfactual delivers 81–93 per cent of the CES gain: variable markups price away 7–19 per cent of what a constant-markup model promises, against a ≈ 20 per cent bite in Liang’s decade. That comparison holds the demand system fixed, and it has to: Section 6.3 shows the correction reverses sign under Dotsey–King and oligopoly-CES calibrations, so what is comparable across the two studies is the Klenow–Willis case, not a specification-free magnitude. The decomposition within the Kimball model attributes most of the remaining market–planner gap (3–8 points) to the diminishing-returns property of Kimball demand rather than to pass-through per se.

That attribution deserves a caution we cannot resolve here. Ren & Zhang (2025) argue that incomplete pass-through, normally read as evidence about the curvature of demand, can instead be produced by monopsony. The mechanism is simple. A monopsonist that cuts employment after a bad productivity shock also drives its own wage down, which partly offsets the rise in marginal cost and weakens its incentive to raise price. Observed pass-through is then below one even when the demand curve alone implies pass-through above one. Our superelasticity is calibrated from the markup–size relationship, so if their mechanism operates here, some of what we read as curvature is the wage response of a monopsonist. This is not a remote possibility on our data: the companion paper finds monopsony in 39 per cent of plant-years on the same panel. We flag it as a limitation in

Section 8 rather than claim the two explanations are separated.

Fourth, gains and calibration co-move. Years of high measured misallocation are years of low margins and high $\hat{\varsigma}$ (compare Tables 7 and 8): the model’s account is internally coherent — intense competition coexisting with badly allocated factors — but the co-movement also flags that part of the year-to-year variation in gains is calibration variation, one more reason to privilege the path’s shape over its points.

5.7 Whose markups? A three-way triangulation

The framework’s markups are demand-side objects. Before trusting them in counterfactuals, Section 4.3’s distribution-level agreement with the production side is pushed to plant level: for the 2017 cross-section we compute, for every plant, the markup under three identification strategies, reported in Table 10 — the calibrated Kimball model; the Atkeson and Burstein (2008) oligopoly-CES model (markup from within-industry market share under nested CES); and the production-side De Loecker–Warzynski estimate from the companion paper, matched for 33,497 of the 37,699 plants.

Table 10: Markups under three identification strategies, 2017 cross-section ($N = 37,699$; 33,497 matched to a production-side estimate). Source: `markup_three_methods_2017.csv`.

	Kimball (demand)	Atkeson–Burstein (oligopoly)	DLW (production)
Mean	1.186	≈ 1.263	≈ 1.30
Std. deviation	0.087	0.013	0.244
Spearman w/ Kimball	1	0.83	0.08
Spearman w/ AB	0.83	1	0.11

Two results, one reassuring and one instructive. The reassuring one: the two *demand-side* estimators, built on different microfoundations (preferences versus conduct), agree strongly on the ranking of plants (Spearman 0.83) — they identify the same plants as the high-markup plants, exactly what one hopes of two theory-consistent measures of one latent object. Both correlate only weakly with the production-side DLW markup (0.08 and 0.11), whose dispersion (0.244) is roughly three times the Kimball and nearly twenty times the Atkeson–Burstein. This is not a defect but confirmation of the methodological point in Liang (2023): the production approach conflates the markup with the plant’s idiosyncratic distortions, so it is far more dispersed and its ranking reflects distortions as much as markups — the very features that make it right for the descriptive market-power portrait of the companion paper and wrong for counterfactual reallocation exercises. The instructive result is the *compression* of the oligopoly markups around the CES value $4.81/3.81 = 1.262$: in Atkeson–Burstein, pricing power requires a large market share, and in the fragmented four-digit industries of the ASI — where the median plant’s share is minuscule — almost every plant is atomistic. Kimball generates size-dependent markups

from preferences without requiring concentration Indian manufacturing does not exhibit, which is exactly why it is the workhorse here. All three estimators agree on the aggregate level (1.19–1.30) and on markups rising with size (the AB markup–share elasticity is +0.003, tiny but precise, $t = 51$).

5.8 The labour-share counterfactual: what reallocation does to the markup-implied labour share

The framework prices reallocation in output. Its distributional question — posed sharply by Zhang and Xia (2022) and by Hsieh et al. (2026) from different directions — is answered by recomputing an aggregate labour share in each of the four equilibria on the 2017 reference year. It is important to be exact about what that object is, because its name invites more than it can deliver. From the model’s first-order condition a plant’s labour share of its own sales is $(1 - \alpha_s)/(\mu_{si} \text{MRPL}_{si})$, and the aggregate is the sales-weighted mean of that across plants and the value-added-weighted mean across industries. No wage and no employment count enters it: w is normalised to one, factor supplies are inelastic, and α_s is fixed from outside the data. In the equalisation counterfactual MRPL is equalised within each industry, so every remaining movement in the aggregate comes from markups and from the sales weights. It is, precisely, the factor share the model’s markups imply, and we call it the markup-implied labour share for that reason.

Two consequences follow and both cut against reading it as a statement about workers’ incomes. It cannot move through the wage, because the wage is fixed by normalisation; and the channel it omits runs the other way, since reallocating factors toward high-productivity plants raises labour’s marginal product and, in a general equilibrium with an upward-sloping labour supply, the wage with it. What the exercise prices is the *pricing* margin: how much of revenue markups retain as profit before any labour-market adjustment. That is a real and under-reported quantity, and it is the one we report.

The result reverses the natural presumption, and the chain has three links worth stating because each is verifiable in the model’s algebra. *Link one: reallocation selects the strong.* Equalising marginal products moves factors toward high-TFPQ plants — under the size–markup gradient, disproportionately plants that already command large shares and charge high markups. *Link two: the winners re-price.* Under (14), a plant that grows faces less elastic demand; the winners of reallocation *raise* their markups. A higher markup is, mechanically, a lower wage-bill share of the plant’s own revenue: the plant retains as profit part of what marginal-cost pricing would deliver to factors. *Link three: composition compounds it.* The counterfactual simultaneously shifts the economy’s weight toward exactly these high-markup plants, so the aggregate share falls by more than any plant’s own change. The CES row isolates the mechanism: with markups frozen, the identical reallocation moves the labour share by -0.12 points — statistically nothing. *The*

Table 11: The aggregate labour share under counterfactuals, 2017 reference year. Every change is measured from one common baseline, the observed allocation: the Kimball baseline reproduces the observed wage-bill share of value added in the estimation sample (0.3427) exactly. The CES model, solved on the same observed wedges, does not, and its misfit is shown as a row of its own. Source: `labor_share_counterfactual_ref2017.csv`; observed share from `16_labor_share_counterfactual.do`.

Equilibrium	Labour share	Δ vs observed
Observed allocation (= Kimball baseline)	0.343	—
Marginal products equalised, Kimball re-pricing	0.316	−2.65 pp
CES on the observed wedges (model misfit)	0.351	+0.78 pp
Marginal products equalised, CES (markups frozen)	0.349	+0.66 pp
of which: CES misfit / reallocation under CES		+0.78 / −0.12 pp
Marginal-cost pricing, actual wedges (planner)	0.430	+8.7 pp
Marginal-cost pricing, equalised (planner)	0.450	+10.8 pp

entire distributional effect operates through the endogenous markup margin. Put on the common observed baseline, the two models end 3.3 points apart after the same equalisation (0.316 against 0.349), and only 0.12 of the CES model’s movement is reallocation; the rest is that model failing to reproduce the observed share before anything is reallocated. Conversely, the planner rows show what does raise labour’s share: marginal-cost pricing lifts it 8.7 points before any reallocation, 10.8 with it. Removing markups is the pro-labour experiment; removing the other wedges, in an economy where markups respond, is not. [Zhang and Xia \(2022\)](#) found for China that markup *heterogeneity* (held fixed) cuts the labour-share gain from liberalisation by a third (+7.4 against +11.4 points); our result sharpens theirs in a diagnosable way — when the markup distribution is allowed to *respond* rather than merely differ, the sign flips. It is the endogeneity, not the heterogeneity, that turns reallocation against labour.

5.9 The size dimension: HKKR diagnostics on the organised sector

Table 12 reports the variance decomposition (21) and the size-neutralising counterfactual, year by year; Table 13 the pooled size gradient itself.

The reading is consistent across all three diagnostics, and Figure 7 plots the first. The size-systematic share of organised-sector misallocation is *small* — 2.2 per cent of within-industry TFPR variance in 2012, rising to 5.6 by 2022 — and the size-neutralising counterfactual yields gains of 1.2–1.9 per cent per year, strikingly close to HKKR’s own 1–4 per cent despite the different sample, and under one per cent of the full-equalisation benchmark computed with the same $\sigma = 3$ machinery. The gradient’s shape (Table 13) explains why: across the small and mid-size bins TFPR is essentially flat (coefficients of +0.02 to +0.06, at most marginally significant), and at the top it turns sharply *negative*

Table 12: Size-dependent wedges, year by year: share of within-industry TFPR variance explained by seven employment-size bins, and the size-neutralising counterfactual gain against the full-equalisation benchmark (both at $\sigma = 3$, HKKR machinery).

Year	Variance decomposition ($\times 100$)			Counterfactual (%)	
	var(TFPR)	size-fitted	share (%)	size-only gain	% of full gain
2012	57.9	1.27	2.2	1.67	0.84
2013	59.1	2.08	3.5	1.78	0.89
2014	54.2	1.90	3.5	1.17	0.61
2015	57.2	2.57	4.5	1.69	0.87
2016	56.7	2.43	4.3	1.37	0.72
2017	54.9	3.04	5.5	1.55	0.83
2018	50.1	2.54	5.1	1.31	0.73
2019	49.4	2.75	5.6	1.90	1.04
2020	49.4	2.57	5.2	1.57	0.87
2021	47.3	2.34	4.9	1.29	0.72
2022	46.3	2.59	5.6	1.34	0.76

Table 13: The TFPR size gradient, pooled 2012–2022: revenue-weighted regression of within-industry log-TFPR deviation on employment-size bins (base: smallest bin), NIC-4 $\times$ year effects. Bin 7 = 500+ workers.

	Bin 2	Bin 3	Bin 4	Bin 5	Bin 6	Bin 7
Coefficient	+0.016	+0.055	+0.041	-0.015	-0.090	-0.337
(s.e.)	(0.034)	(0.032)	(0.031)	(0.031)	(0.031)	(0.031)

— plants with 500+ workers have TFPR 34 log points below the smallest bin. Whatever taxes size in India, it is not visible as a rising wedge schedule *within* the organised sector; the largest factories behave, if anything, as wedge-favoured (consistent with better credit access — recall capital wedges dominate — and with the residual market-power component of TFPR).

Their motivating facts split instructively at the formality boundary (Table 14): the organised-sector *wage* gradient is even steeper than theirs — average wages at 500+-worker plants exceed the smallest plants’ by roughly 113 log points, against their ≈ 60 per cent — but the white-collar employment share *falls* with plant size, from 27.8 per cent in the smallest bin to 19.8 at 500+, the opposite of their education gradient, which rises from 30 to over 60 per cent once informal micro-enterprises are included.

Table 14: HKKR’s motivating gradients inside the organised sector: wage premium by size bin (log points, relative to bin 1) and the white-collar (non-production) employment share.

	Bin 1	Bin 2	Bin 3	Bin 4	Bin 5	Bin 6	Bin 7
Wage premium	0	0.234	0.397	0.589	0.722	0.871	1.126
	—	(0.031)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
White-collar share	27.8	26.5	27.4	25.9	23.1	21.7	19.8
Plants (000s)	56	65	88	97	69	177	53

The synthesis matters for policy. Size-dependent misallocation — and with it HKKR’s efficiency–inequality trade-off through the skill premium — is concentrated at the *formal–informal margin*, which lies below the ASI’s coverage floor. Within organised manufacturing the size margin is thin, and reallocation toward larger plants would if anything tilt labour demand toward *production* workers. The large gains of Table 8 are therefore within-formal-sector gains, not gains from dismantling size-dependent policy; and the distributional caution this paper carries (Table 11) is the organised-sector counterpart — operating through variable markups rather than skill composition — of HKKR’s finding that efficiency-enhancing reallocation need not benefit workers.

5.10 Geography: states, regions, and the rural–urban margin

Misallocation has a geography, and it is not the geography of market power. Table 15 summarises by region (the full 28-state panel is Appendix C; Figure 8 maps it), and Table 16 tracks the rural–urban split by year.

Three patterns, with Figure 9 ranking the states the map shades. **First, the state map.** Dispersion is lowest in the established north-western industrial belt — Punjab (49.1) and Haryana (49.4) by a wide margin, then Gujarat (56.5) and Tamil Nadu (60.4) — and highest in the east and peninsula: Odisha (77.6), Andhra Pradesh (76.5), West Bengal (75.9), Kerala (72.9), Jharkhand (71.7). Plants in the most misallocated states are roughly

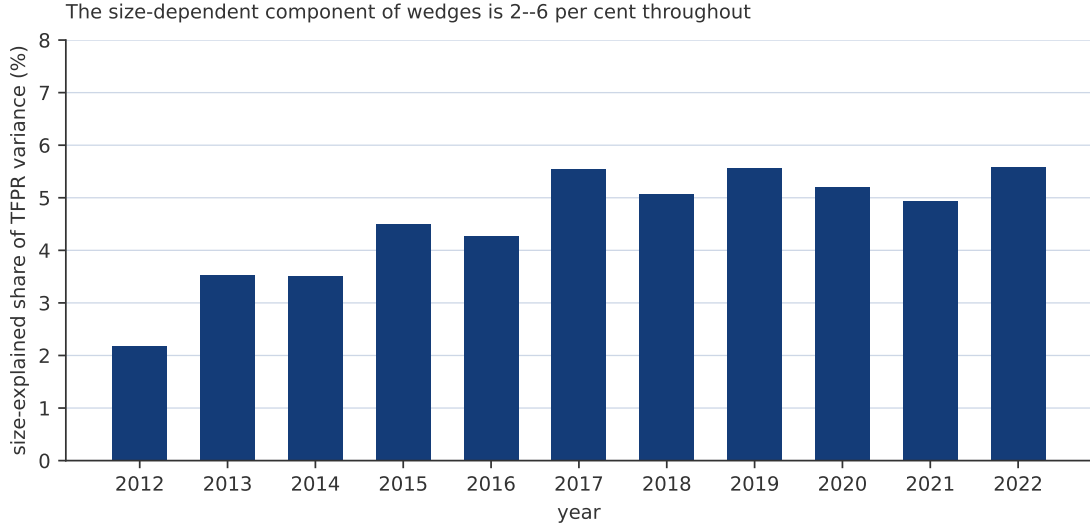


Figure 7: Share of within-industry TFPR variance explained by seven employment-size bins, by year. The size-dependent component of wedges stays between two and six per cent throughout, with no upward trend.

Table 15: Misallocation by region, pooled 2012–2022: within-industry variance ($\times 100$) of TFPR and MRPL; median markup and monopsony incidence from the companion paper for comparison.

Region	N	var TFPR	var MRPL	med μ	$\Pr(\psi > 1)$
East	37,688	71.8	60.5	1.228	0.467
Northeast	20,246	67.3	50.3	1.304	0.367
South	123,491	66.5	56.7	1.233	0.365
Central	20,265	65.5	61.5	1.183	0.566
West	97,714	62.2	55.2	1.221	0.386
North	130,130	59.5	50.3	1.219	0.438

Table 16: Rural and urban misallocation by year: within-industry $\text{var}(\log \text{TFPR}) \times 100$.

Year	Rural	Urban	Gap (R–U)
2012	73.1	72.5	+0.6
2013	73.2	70.6	+2.6
2014	70.1	67.5	+2.7
2015	68.6	67.4	+1.2
2016	65.9	63.3	+2.6
2017	63.7	63.2	+0.5
2018	59.5	57.5	+2.0
2019	60.1	59.4	+0.7
2020	60.6	59.3	+1.3
2021	58.3	58.0	+0.3
2022	58.9	58.0	+0.9

Misallocation by state (var log TFPR x100), 2012-2022

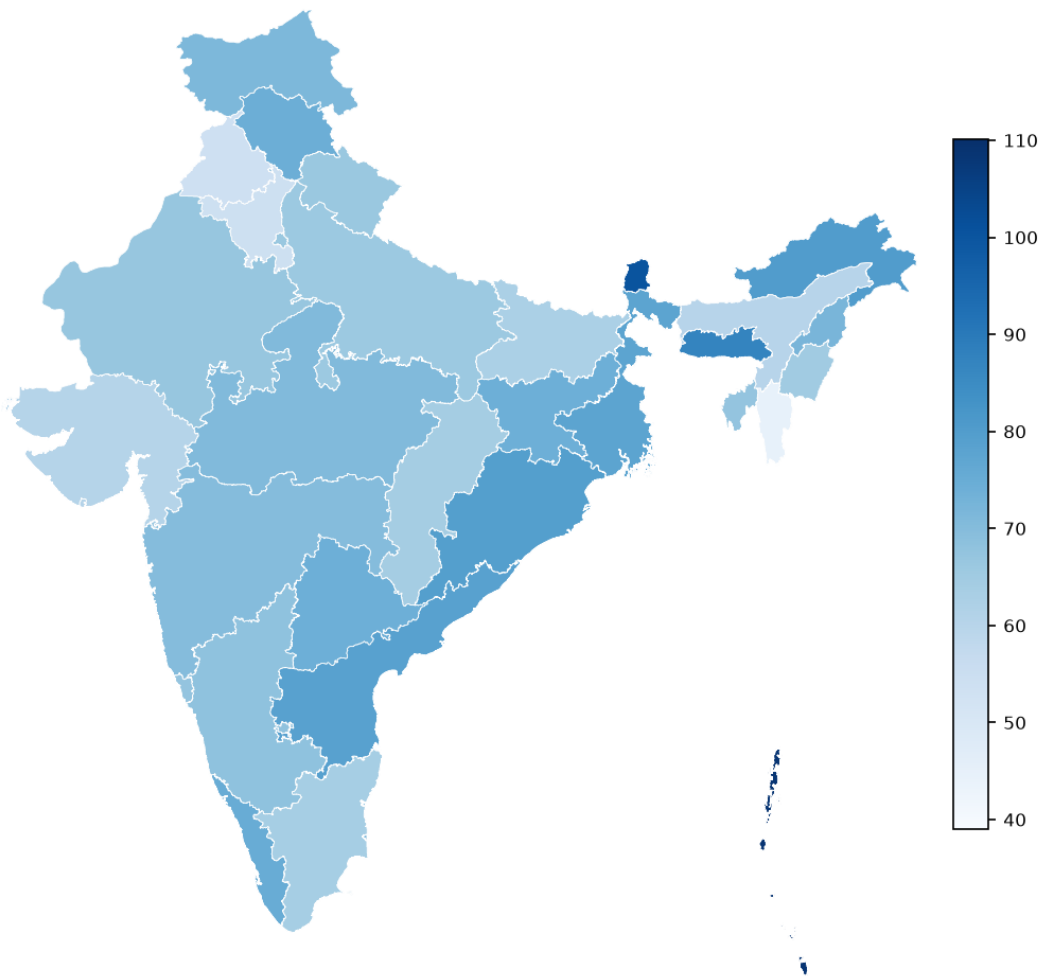


Figure 8: Within-industry TFPR dispersion by state, pooled 2012–2022. Darker = more misallocated.

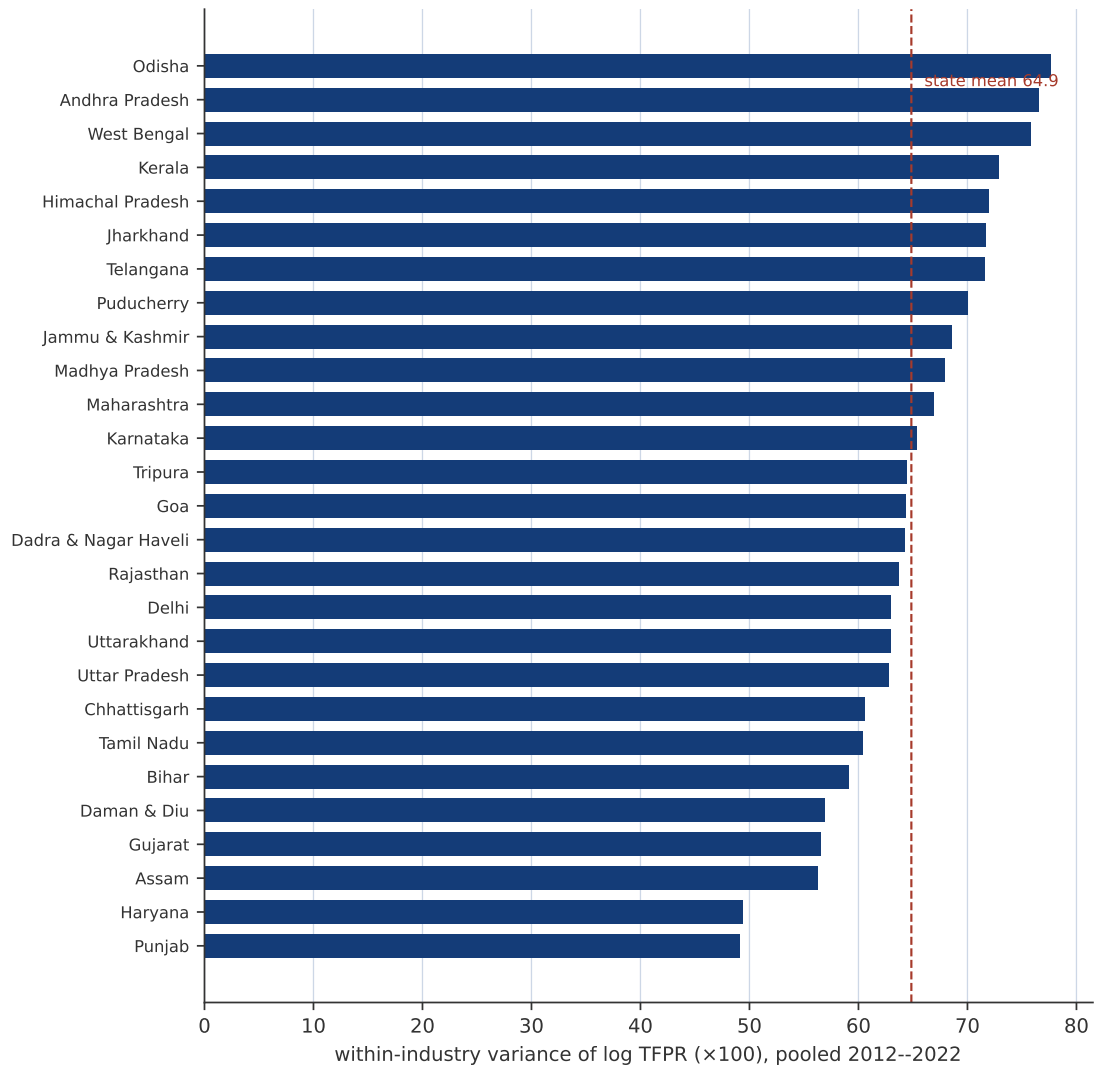


Figure 9: The same quantity ranked: within-industry variance of log TFPR by state, pooled 2012–2022. The dashed rule is the unweighted mean across states.

sixty per cent more misallocated, on this metric, than in Punjab or Haryana. The capital column dominates the labour column in *every* state — the factor anatomy of Section 5.1 is not a composition effect. **Second, the cross-margin comparison.** Set against the companion paper’s geography, the two maps differ in their headline: monopsony peaks in the central-eastern mineral belt, misallocation in the east and peninsula; Punjab and Haryana are efficient on both margins, West Bengal poor on both. On the aggregate measure, then, market power and misallocation are correlated neighbours rather than the same variable, which is what the variance accounting (markups ≈ 1.6 per cent of TFPR dispersion) already implied at plant level. That comparison is nevertheless made against the wrong object, because the aggregate is dominated by the capital wedge. Section 5.11 makes it against the right one, and the answer changes. **Third, the rural–urban margin is thin.** Pooled rural dispersion exceeds urban by little (64.4 vs 63.1), and the by-year series converge over the decade. The strong spatial dimension of Indian misallocation is the *state*, not the village–town boundary — in sharp contrast to monopsony, where the rural–urban gap is the first-order spatial fact. Whatever drives factor-market wedges apart operates at the level of state institutions (credit depth, land administration, power reliability), not settlement type.

5.11 Does labour-market power sit where labour is misallocated?

Theory says this comparison is the one to make. [Trottner \(2022\)](#) builds a heterogeneous-firm model with endogenously variable markdowns and markups, and shows that monopsony power is not in itself an aggregate distortion when markdowns are homogeneous. What distorts is their *dispersion*. He also shows that the misallocation caused by monopsony cannot be assessed without accounting for firms’ product-market power at the same time, and conversely. That is the formal justification for measuring both wedges on the same plants, which is what this paper and its companion do between them, and it is why the object to correlate with monopsony is labour-wedge dispersion rather than the level of anyone’s markdown.

The comparison just made sets the companion paper’s monopsony geography against *total* TFPR dispersion. That is the natural first comparison and it is the wrong one, for a reason internal to this paper: total dispersion is dominated by the capital wedge, which is half again the labour wedge in every year and every state (Section 5.1). A labour-market friction should not be expected to show up in an aggregate that capital controls. The sharper test decomposes.

We therefore match the two papers state by state. The companion paper reports monopsony incidence $\Pr(\psi > 1)$ for the 21 states that clear its 5,000 plant-year ranking threshold; this paper reports the within-industry variance of log MRPL and log MRPK deviations for the same states, built from the same panel, the same window and the same

state coding. Both are aggregates of the same plants, but they are constructed by entirely different routes: the monopsony measure comes from a production-function markdown of the kind [Brooks et al. \(2021\)](#) introduced for India and China, the ratio of two elasticity-to-share ratios, while the wedge dispersions come from the Hsieh–Klenow accounting and never use an estimated production function at all. Table 17 reports the correlations.

Table 17: Monopsony incidence (the companion paper) against factor-wedge dispersion (this paper), across the 21 states present in both, pooled 2012–2022. Pearson r with its two-sided p -value on 19 degrees of freedom, and Spearman ρ .

Monopsony incidence $\Pr(\psi > 1)$ versus	r	p	ρ
Labour-wedge dispersion, var log MRPL	+0.542	0.011	+0.639
Capital-wedge dispersion, var log MRPK	+0.114	0.622	+0.099
Total dispersion, var log TFPR	+0.293	0.197	+0.338
Capital/labour wedge-variance ratio	−0.528	0.014	−0.503

The pattern is clean and it discriminates. Monopsony incidence is strongly and significantly associated with the dispersion of the *labour* wedge ($r = +0.54$, $p = 0.011$; $\rho = +0.64$) and essentially unrelated to the dispersion of the *capital* wedge ($r = +0.11$, $p = 0.62$). The correlation with total dispersion, $+0.29$, is not significant and is simply the average of a real relationship and a null one, diluted by the capital wedge’s dominance of the aggregate. Figure 10 shows both panels.

Twenty-one states is a small sample, so the obvious question is whether one of them carries the result. It does not. Dropping each state in turn and recomputing, the labour-wedge correlation stays inside $[+0.489, +0.610]$ against its full-sample $+0.542$, and its sign is unchanged in all twenty-one leave-one-out samples; the widest single move is Bihar, worth -0.053 . The capital-wedge correlation stays inside $[+0.006, +0.236]$ — that is, indistinguishable from zero however the sample is cut. The rank correlations in Table 17 tell the same story without the functional form: $\rho = +0.64$ on labour against $+0.10$ on capital. None of this makes twenty-one observations into a large sample, and none of it is causal — the two measures are built from the same plants, so a common state-level determinant of both is entirely possible and is not ruled out by any test here. What the diagnostics do establish is that the labour–capital contrast is a property of the panel rather than of one or two influential states.

A confound worth testing, and clearing. The obvious objection is that this is a capital-intensity association wearing a labour-market costume. Labour-intensive states would plausibly have both more employer wage-setting power and more dispersion in the labour wedge, and the pattern in Table 17 would then be composition rather than mechanism. The objection has real force, because capital intensity *is* strongly related to

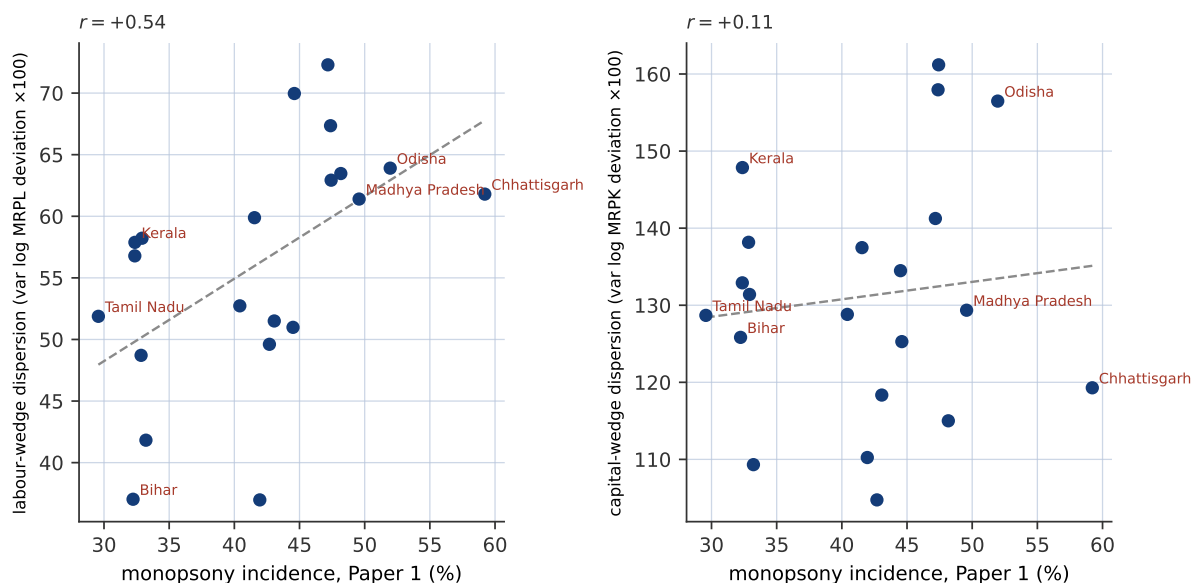


Figure 10: Monopsony incidence from the companion paper against this paper’s factor-wedge dispersion, 21 states. Left: the labour wedge. Right: the capital wedge. Dashed lines are ordinary least-squares fits. The two measures are built by different methods from the same plants; they agree on labour and not on capital.

monopsony incidence across these states: the correlation with log real capital per person engaged is $+0.59$ ($p = 0.002$), and with log real capital per rupee of wage bill $+0.60$ ($p = 0.001$).

It does not survive contact with the data. Holding capital per rupee of wage bill fixed — the measure in the model’s own units, since labour enters as the wage bill — the partial correlation of monopsony incidence with labour-wedge dispersion is $+0.48$ ($p = 0.019$), against a raw $+0.54$; on capital per person engaged it is $+0.40$ ($p = 0.068$). The capital-wedge association, null to begin with at $+0.11$, stays null on both: -0.02 and -0.05 . The *gap* between the two, which is what the finding actually rests on, is 0.43 raw and 0.50 once capital intensity is held fixed. It widens rather than narrows.

One clarification, because the state table invites the confusion: the capital/labour ratio reported in Table 17 is the ratio of wedge *variances*, $\text{var ln MRPK} / \text{var ln MRPL}$, not an input ratio. Conditioning on that would remove part of the variation under study rather than a confound, so the test above builds capital intensity from the estimation panel instead (Appendix code R15, R16). With twenty-one states these partial correlations are not precise, and we do not present them as such; what they establish is that the discrimination between the two factor markets is not an artefact of how capital-intensive a state’s manufacturing happens to be.

The pairing also speaks to a live disagreement about size. Ren & Zhang (2025) find on US data that larger firms have *lower* product-market power and *higher* labour-market power, whereas the companion paper finds markups rising with size in India. The two need not conflict, since the samples are public US firms and Indian registered factories

and the size ranges barely overlap, but we record the contrast rather than assume it away. What both agree on is the sign on the labour side: market power over workers rises with employer size.

Three things follow. First, this is mutual validation of a kind neither paper could supply alone. Two measurement strategies with almost no machinery in common — one estimating production functions, one differencing revenue productivities against industry aggregates — identify the same states as having distorted labour markets. That agreement is not built in; had the markdown been an artefact of the estimator, it would have no reason to track a wedge dispersion computed without one.

Second, it locates monopsony correctly within the allocative accounting. The companion paper shows employer wage-setting power is pervasive; this paper shows the labour wedge is the smaller half of measured misallocation. Both are true, and the correlation explains how: monopsony is a first-order fact about the labour wedge and close to irrelevant to the capital wedge, which is the part that dominates the aggregate. The negative correlation with the capital-to-labour variance ratio ($r = -0.53$, $p = 0.014$) states the same thing from the other side — high-monopsony states are not states with more distortion overall, but states whose distortion is unusually *labour*-weighted.

Third, it sharpens the policy reading of Section 7. A state with a labour-weighted distortion profile and a state with a capital-weighted one need different instruments, and the two groups are identifiable in advance from either paper’s measure. The result also has a direct antecedent. [Mohammad et al. \(2021\)](#) exploit micro-level variation across Indian states and find that states with less rigid labour markets have less misallocation, with the effect interacting with informality. Their labour-market rigidity is our labour wedge seen from the policy side, and their cross-state design is the one our correlation describes in cross-section. That two literatures reach the same state ordering by different routes is worth more than either alone.

It also connects the measurement to a design literature. The move from monopsony estimates to instruments is surveyed by [Azar & Marinescu \(2024\)](#), whose policy menu — minimum wages, restraints on mobility-suppressing contract terms, competition scrutiny of labour markets — is aimed at exactly the margin our correlation isolates. India’s states differ in employment protection, which has long been used to identify effects on manufacturing outcomes ([Besley & Burgess, 2004](#)), as have size-contingent thresholds in central statutes ([Bertrand et al., 2021](#)). That variation acts on the labour margin, which is the margin our correlation says carries the monopsony signal. What the pairing cannot do is establish direction: employer wage-setting power and labour-wedge dispersion are two views of the same cross-section, and nothing here identifies which is cause. We report the association because it is strong, because it is specific to the factor market theory says it should be specific to, and because a null result on capital is what makes it informative.

6 Robustness and sensitivity

This section implements the [Zhang and Xia \(2022\)](#) programme: vary every maintained parameter, report every level shift, and ask which conclusions move.

6.1 Domestic capital elasticities: a degeneracy worth reporting

The baseline imports U.S. capital elasticities. The natural domestic alternative — industry cost shares, $\alpha_s = 1 - \sum wL / \sum VA$ pooled over 2012–2022 — lies in $[0.37, 0.93]$ with mean 0.70, systematically above the U.S. mean of 0.57, because Indian wage-bill shares of value added are low. Implementing it exposes a structural degeneracy that we report as a finding: with cost-share elasticities the aggregate Lerner markup is ≈ 0.89 *by construction* — the shares absorb the average markup, pushing measured factor payments above value added — so the implied elasticity target in (16) is negative and the demand system cannot be calibrated at all. This is HK’s original argument for external elasticities realised exactly in our data. The economically meaningful experiment (following Zhang–Xia’s own separation) holds the demand system at the baseline calibration and varies only α_s : measured TFPR dispersion rises from 66.3 to 87.4; the markup share of that dispersion falls from 1.6 to 0.4 per cent; the 2017 Kimball market gain rises from 175 to 220 per cent; and the labour-share counterfactual is dampened but unchanged in sign (-0.94 against -2.65 points). Every qualitative conclusion survives; every level moves, exactly as Zhang–Xia warn — and in the direction their analysis predicts, since higher capital shares load more dispersion onto the capital wedge. An attempted third route — converting the companion paper’s estimated production functions to value-added form, $\alpha_s = \theta_K / (\theta_K + \theta_L)$ — fails for a known reason: gross-output control-function estimates of book-capital elasticities are attenuated, and the conversion yields $\alpha_s \in [-0.22, 0.43]$ including negative values, unusable in a framework requiring $\alpha_s \in (0, 1)$.

6.2 Book-value capital and the vintage channel

Capital here is the nominal closing net book value of fixed assets, and labour is the current-year wage bill. That asymmetry deserves its own treatment, because the paper’s headline anatomy — capital-wedge dispersion half again labour-wedge dispersion in every year and every state — compares a well-measured current flow against a stock accumulated at historical cost over the plant’s lifetime.

The justification this literature gives for nominal values, which Section 3 repeats, is that every comparison is made inside an industry-year cell so industry-year price levels cancel. That is correct for a common deflator and it does not reach the problem. Two plants in the same NIC-4 cell in the same year, with identical real capital but different investment vintages, carry different book values, and the gap grows with cumulative

inflation over the spread of their vintages. This is a *within*-cell, cross-plant discrepancy; no industry-year index removes it. A plant that bought its machines fifteen years ago shows understated K , therefore overstated $\text{MRPK} = Py/K$, and reads as capital-starved when it is merely old.

The panel carries the gross book stock as well as the net, so the undepreciated fraction $K^{\text{net}}/K^{\text{gross}}$ is available as a vintage proxy. It is imperfect — it moves with depreciation policy and asset mix too — but it is the vintage information these data contain. Table 18 regresses each wedge on it within industry-year.

Table 18: Each within-industry-year wedge regressed on the undepreciated fraction of the book capital stock. Estimation sample reproduced exactly (429,534 plant-years); the proxy is usable on the 237,507 with a positive gross stock. Source: `vintage_check.csv`, `vintage_deciles.csv`.

Wedge	Within-cell variance	Absorbed by vintage	Coefficient
Capital, $\ln \text{MRPK}$	1.308	16.4%	−2.124
Labour, $\ln \text{MRPL}$	0.633	2.6%	+0.585
Revenue productivity	0.680	6.3%	−0.956

The channel is real and asymmetric exactly as the concern predicts. Vintage absorbs six times more of the capital wedge than of the labour wedge, and with the right sign: a lower undepreciated fraction, meaning an older asset base, raises the measured capital wedge. The profile is monotone across all ten vintage deciles, from a mean capital wedge of +1.07 in the oldest decile, where the net-to-gross ratio averages 0.20, to −0.49 in the newest, where it averages 0.90. That is a spread of 1.56 log points within industry-year against a wedge standard deviation of 1.14. The labour wedge moves a third as far and in the opposite direction.

The headline survives it. Removing the vintage-fitted component moves the capital-to-labour dispersion ratio from 1.447 to 1.340. Capital-wedge dispersion is still a third larger than labour’s, and “capital carries the misallocation” stands.

So we built the correction rather than bounding it. The panel carries actual additions to fixed assets and the machinery deflator, so a perpetual-inventory stock is available: $K_t = (1 - \delta)K_{t-1} + I_t$ in constant prices, with $\delta = 0.10$, seeded on the plant’s first observed real book value and re-seeded at gaps in its run. The series behaves as it should. Because it is seeded at book value it starts there by construction, and the median ratio of perpetual-inventory to book capital then rises monotonically with tenure, from 1.00 in a plant’s first observed year to 1.44 in its eleventh. Book value understates real capital, the understatement grows with the age of the asset base, and the correction recovers it.

Table 19 re-runs the two statements that depend on the capital measure. Neither moves.

The dispersion barely moves. Replacing book capital with a perpetual-inventory

Table 19: The capital wedge under book value and under perpetual-inventory capital. All wedges are within-industry-year, trimmed at the 1 per cent tails by year as elsewhere. 320,980 plant-years, the subset on which every capital variant is positive. The last row seeds the recursion on the gross rather than the net book stock. Source: `pim_wedges.csv`, `pim_persistence.csv`.

Capital measure	s.d. capital wedge	vs labour	ρ at 1 yr	ρ at 5 yr
Net book value (baseline)	1.099	1.509	0.776	0.535
Perpetual inventory, $\delta = 0.10$	1.069	1.467	0.797	0.552
$\delta = 0.05$	1.073	1.473	—	—
$\delta = 0.15$	1.072	1.471	—	—
seeded on gross book	1.442	1.980	0.843	0.585

stock reduces the standard deviation of the capital wedge by 2.7 per cent, from 1.099 to 1.069, and the ratio to the labour wedge from 1.509 to 1.467. The result is insensitive to the depreciation rate: at $\delta = 0.05$ and $\delta = 0.15$ the dispersion is 1.073 and 1.072. Seeding on the gross rather than the net stock moves it the other way, to 1.442. Capital wedges dominate labour wedges under every capital measure we can construct.

And the persistence rises rather than falls, which settles the question against us. If the durability of the capital wedge were an artefact of a permanent accounting characteristic, correcting the accounting would reduce it. It does the opposite: the one-year autocorrelation goes from 0.776 under book value to 0.797 under perpetual inventory and 0.843 under the gross-seeded variant, and the five-year figure from 0.535 to 0.552 and 0.585. The vintage channel is not what makes the capital wedge persistent.

Why, then, did the undepreciated fraction absorb 16.4 per cent of the variance? Because it is not a clean measure of vintage. A plant with a heavily written-down asset base is also a plant that has not invested, and not investing is exactly what a credit-constrained plant does. The proxy was picking up the distortion along with the accounting, which is the endogeneity we flagged when we bounded it; the perpetual-inventory series separates the two and shows the pure measurement component is small. The honest summary is that a real concern was raised, tested properly, and did not survive: the capital results in this paper do not rest on the book-value convention.

One limitation of the exercise remains. The recursion is seeded on book value, so the earliest observation of each plant still carries the book-value problem in full and the correction only accumulates with tenure. A panel that observed plants from birth would correct more. Given that the measured effect is 2.7 per cent on a series where the median correction reaches 44 per cent by year eleven, we do not think a longer panel would overturn the conclusion, but we cannot demonstrate that here.

6.3 The demand system

Table 20 re-solves the 2017 counterfactuals under three alternative demand systems, each recalibrated to the same targets.

Table 20: 2017 counterfactual gains under alternative demand systems (all recalibrated to the 2017 targets; no capped sectors). Comparison, Liang (2023) Table 5 (2004): translog 86.7, Dotsey–King 109.0, oligopoly-CES 118.1, Kimball baseline 105.1.

Demand system	Market	Planner	CES	Market/CES
Kimball–Klenow–Willis (baseline)	175.1	178.7	188.6	0.93
Translog (Feenstra, 2003)	171.0	197.9	188.6	0.91
Dotsey–King (2005)	201.8	211.1	188.6	1.07
Oligopoly-CES (Atkeson and Burstein, 2008)	205.6	209.8	188.6	1.09

The pattern matches Liang’s qualitatively and refines the interpretation. Under translog demand, whose elasticity falls *faster* with size than Klenow–Willis, the market gain sits below its own planner gain by the largest margin of any specification (171 vs 198) — faster-declining elasticities strengthen the incomplete-pass-through channel exactly as theory predicts. Under Dotsey–King and the oligopoly model, whose elasticities fall more slowly, variable markups bite so much less that the market gains exceed the CES benchmark outright — a reversal of the baseline’s sign, not a weakening of its magnitude. It has to be said plainly that this table does not support a claim of robustness, and the paper does not make one. Under the two systems whose elasticity falls faster with size — Klenow–Willis and translog — variable markups reduce the gain, by 7 and 9 per cent respectively. Under the two whose elasticity falls more slowly they *raise* it, by 7 and 9 per cent. The *sign* of the markup correction is therefore a property of the demand system, not a finding about Indian manufacturing, and any reader who takes the headline 7–19 per cent as a specification-free magnitude is being misled.

What survives specification choice is narrower and worth stating exactly. The gap between the market and planner equilibria — incomplete pass-through, the wedge between what a reallocating economy delivers and what marginal-cost pricing would deliver — is negative in all four rows, and widest under translog, where elasticities fall fastest (171 against 198). That is the mechanism the superelasticity governs, and it is signed by theory rather than by calibration. The comparison against CES is a different object: it asks whether variable markups raise or lower the gain relative to a model that has no markup dispersion at all, and the answer depends on whether the calibrated elasticity schedule is steeper or flatter than the CES benchmark it is being measured against. Our preferred estimate $\widehat{\varepsilon/\varsigma} = 0.183$ is what puts the Kimball row on the reducing side; a demand system calibrated to a flatter schedule sits on the other. The honest summary is that the markup tax reported in Section 5.6 is conditional on the Klenow–Willis specification and on that estimate, and the paper reports its range under alternatives here rather than burying it.

(One caveat recorded for completeness: Liang’s translog calibration disciplines only the average elasticity, leaving the implied superelasticity free — 1.43 here against the 0.183 target — so the translog row is not a like-for-like superelasticity match; this follows her code exactly and is flagged rather than “corrected”.)

6.4 Inference: what the estimates are worth

Every number reported so far is a point estimate. The counterfactual gains are constructed objects — they depend on a demand elasticity estimated from the sample, a superelasticity estimated from a regression on the sample, and a fixed point solved on the sample — so treating them as data would understate their uncertainty in a way that no amount of specification robustness can repair. Section 6 varies the maintained assumptions; this subsection varies the sample.

We bootstrap the entire pipeline. In each replication we draw plants with replacement, give each drawn copy its own identity so a plant drawn twice counts as two units in every fixed effect and every sectoral aggregate, and then re-run everything the sample touches: the Hsieh–Klenow cleaning, including the cell-size rule and the one per cent tail trims that are themselves sample-dependent; the calibration of the demand elasticity and of the superelasticity; the outer and inner secants that solve the Kimball fixed point; and both counterfactuals. Nothing is held at its full-sample value. As a check that the procedure wraps the published estimator rather than a variant of it, replication zero — the full sample with no resampling — reproduces the published estimates to seven significant figures.

Table 21: Plant-clustered bootstrap inference, 172 converged replications of 200 attempted. Each replication redraws plants with replacement, re-runs the Hsieh–Klenow cleaning including its sample-dependent trims, re-estimates the demand elasticity and the superelasticity, and re-solves the Kimball fixed point and both counterfactuals. Point estimates are the full-sample values. Intervals are percentile intervals and are conditional on convergence; Section 6.4 explains why that qualification matters.

Statistic	Point	Bootstrap s.e.	95% interval
Kimball market gain (%)	175.054	16.601	[129.46, 194.13]
Kimball planner gain (%)	178.675	16.435	[133.89, 197.62]
CES gain (%)	188.578	19.274	[133.58, 208.05]
Market/CES ratio (%)	92.828	2.557	[91.07, 100.32]
Markup tax on the cure (%)	7.172	2.557	[-0.32, 8.93]
Calibrated $\hat{\varsigma}$	5.340	0.409	[4.29, 5.77]
Superelasticity	0.183	0.005	[0.17, 0.18]
s.d. log TFPR deviation	0.837	0.007	[0.83, 0.86]
s.d. log MRPK deviation	1.208	0.009	[1.20, 1.23]
s.d. log MRPL deviation	0.774	0.008	[0.77, 0.79]

Three readings follow, and they do not all favour the paper.

The dispersion facts are very precisely estimated. The standard error on the standard deviation of the log TFPR deviation is 0.0074 against a point estimate of 0.837, about 0.9 per cent of its value; the capital and labour wedge dispersions are estimated to 0.8 and 1.0 per cent respectively. The anatomy of Section 5.1 — capital wedges half again the size of labour wedges, and both falling — is not in any doubt from sampling variation. Neither is the persistence result, which rests on the same plant-level moments.

The counterfactual gains are not. The Kimball market gain carries a bootstrap standard error of 16.6 points on a point estimate of 175.1, with a 95 per cent interval of [129, 194]. That is a wide interval, and it is wide for a reason worth naming: the gain inherits the sampling uncertainty in the calibrated $\hat{\varsigma}$, whose own standard error is 0.41 on 5.34, and the mapping from ς to the gain is steep. The paper’s qualitative claim — that misallocation in Indian manufacturing is large — survives comfortably, since the whole interval lies far above zero. Any reading that leans on the second or third digit of a gain does not.

The markup tax is the fragile one. At the 2017 reference year the Kimball market counterfactual delivers 92.8 per cent of the CES gain, so the tax is 7.2 per cent, with a standard error of 2.6 and an interval of [-0.3, 8.9]. That interval includes zero, so at this reference year the markup tax is not distinguishable from no tax at all at conventional levels. Two things should be said about that. The first is that 2017 is the year least favourable to this finding: across reference years the market/CES ratio runs from 0.81 to 0.93, and 2017 sits at the top of that range, so the tax being bootstrapped is the smallest one the decade offers. The second is that the finding is a statement about a *range across years*, 7 to 19 per cent, not about one year’s point estimate, and the bootstrap speaks only to the latter. We would not want a reader to take 7–19 per cent as though each end carried the precision of the dispersion statistics. It does not.

A selection problem in the bootstrap itself, and what it does to the intervals. Of 200 replications attempted, 172 converged and 28 — 14.0 per cent — were discarded because the counterfactual solver failed to reach its tolerance within a bounded number of iterations. This is not a computational nuisance to be reported in a footnote. It is the same phenomenon as the paper’s own 2019 boundary case (Section 6.7): a cross-section whose calibration lands at a high demand elasticity makes the Kimball fixed point hard, and sometimes impossible, to solve.

The discarded draws are therefore not a random subset. Their calibrated $\hat{\varsigma}$ has a median of 6.25 against 5.13 among the draws that converged, and their maximum reaches 7.35. The replications that survive systematically under-represent the upper tail of ς , and since the gain rises with ς , the intervals above are conditional on convergence and are, at the top end, too narrow. We report them as such rather than presenting them as unconditional confidence intervals. It is possible to go further than the concession,

however, and bound the damage rather than merely flag it. Because the gain is increasing in ς and the failures sit in the upper tail of ς , assigning every failed draw the *largest* gain observed among the converged ones yields a valid upper bound on the top of the interval, and assigning the smallest yields the matching lower bound; the truth lies between, and the published conditional interval necessarily does too. On that construction the 2017 Kimball market gain moves from a conditional [129.5, 194.1] to a worst-case [115.5, 198.8], and the markup tax from $[-0.3, 8.9]$ to $[-5.3, 11.7]$. The qualitative readings are unchanged — the gain remains far from zero under the most adverse assignment, and the markup tax remains indistinguishable from zero at this reference year — but a reader should take the wider pair as the honest summary of what 200 attempted draws support. One further feature of the draw distribution belongs here rather than in a footnote: the bootstrap mean of the markup tax, 4.3 per cent, sits below the full-sample point estimate of 7.2, so the resampling distribution is not centred on the statistic it is bootstrapping. That is a further reason to treat this particular quantity as the fragile one. The complete draw-level record — all 200 attempts, converged and failed, with their calibrated parameters, both counterfactual gains and solver outcome — is shipped as `Results/p2_boot_drawlog.csv` so that these statements can be checked rather than taken on trust. The honest summary is that the uncertainty in these counterfactuals is at least as large as the table says and probably larger on the upside, and that a model class with a solvable domain this ragged is itself a finding about the framework — one that a single-cross-section study, which either solves or does not, has no way of seeing.

6.5 Accounting, demand system, and elasticities: where they are treated

The two remaining maintained choices are addressed elsewhere because they turned out to be results rather than checks. The gross-output versus value-added accounting is Section 5.2, where it belongs: it moves the measured level by a factor of seven and is a finding about the literature, not a sensitivity of ours. The domestic-elasticity degeneracy is Section 6.1 above.

6.6 Survey weights

Every statistic in the main text treats each surveyed plant as one observation. The ASI is a census of large units and a sample of the rest, and each plant carries a multiplier saying how many registered factories it stands for (median 1, mean 3.1, maximum 60). Weighting by it changes the object from the surveyed sample to the population of registered factories, so it is worth knowing whether any comparison depends on the choice. None does. Two checks reproduce published numbers before anything is varied: the unweighted within-industry dispersion series matches Table 5 to 5×10^{-15} , and the multiplier-weighted $\sigma = 3$

full-equalisation gains of Section 5.9 match to 5×10^{-6} (38_survey_weighted_p2.py, survey_weighted_p2.csv).

With the multipliers, the within-industry standard deviation of log TFPR moves by at most 0.015 in any year, capital wedges remain wider than labour wedges in all eleven years, and dispersion falls by 9.9 per cent between 2012 and 2022 against 10.3 per cent unweighted. The $\sigma = 3$ full-equalisation gain is 4.7 to 8.7 per cent higher with weights, and the two annual paths have a rank correlation of 0.99. The Kimball gains are not recomputed on weights, because that would require reweighting the equilibrium solver itself; on the evidence of the benchmark, weighting would raise their level a little and leave their path unchanged, but we have not shown this.

6.7 The 2019 boundary case and other checks

The 2019 calibration — elasticity target 6.27 from the decade’s thinnest aggregate margin — converges only at $\hat{\varsigma} = 7.4$ with gains of 270 per cent, far off the neighbouring years; small perturbations of the target move the solution violently, the signature of a model class approaching its solvable boundary (the Klenow–Willis aggregator requires interior relative sizes, and high ς pushes mass toward the boundary). We report 2019 in Table 8 for completeness and exclude it from every statement about the path. The rule is that a year leaves the path when its calibration lands at the edge of the aggregator’s solvable domain, and 2019 is the only year that does. The rule was written after we saw 2019, not before, so we also give the inclusive reading: with 2019 the range of annual gains is 121–270 per cent rather than 121–184, and the path acquires a spike in 2019. Excluding tobacco — the $\alpha_s = 0.95$ outlier — from the 2017 model moves the gain from 175.1 to 158.3 per cent with $\hat{\varsigma} 4.94$: a visible level effect confined to one division, no change in any comparison. Finally, the trims: all results use the literature’s 1 per cent trims of TFPR and TFPQ deviations; the measured decline in dispersion is present in the untrimmed data as well, so the trend does not ride on outlier handling.

7 Interpretation: market power and misallocation, jointly read

Read together with the companion paper, the two margins of market power and the misallocation accounting form one picture with a clear division of labour. *Product-market power* is modest, tightly dispersed, and a small component of measured misallocation (1.6 per cent of TFPR variance) — but because it is size-dependent and endogenous, it governs how much of the reallocation dividend is attainable (81–93 per cent) and who receives it (not labour: −2.65 points of wage share). *Labour-market power* is pervasive at plant level, yet the labour *wedge dispersion* it feeds is dominated by capital wedges

in every year and state — monopsony’s cost is borne first by workers (the companion paper’s incidence and welfare results) and only second by aggregate allocation. *Residual wedges* — credit, land, regulation, enforcement, whose geography is state-institutional rather than rural–urban — carry the overwhelming share of measured misallocation, and their decade-long decline is the sample’s good news.

Three features of the Indian policy setting sharpen what that division of labour implies. The instruments currently in the field are output- and investment-side: Make in India and the Production Linked Incentive schemes commit Rs. 1.97 lakh crore across fourteen sectors to capacity, exports and competitiveness (PIB, 2025), and NITI Aayog (2026) assesses twelve priority sectors for a manufacturing hub on market potential, infrastructure, technology readiness and employment potential. Nothing in that toolkit acts on the pricing margin, which our planner counterfactual scores as the single most pro-labour intervention available. Meanwhile the diagnosis in the wider Indian debate is that growth has been unusually capital- and skill-intensive for the country’s stage of development, with the labour-intensive segments growing slowly (Bishnoi et al., 2024) and output and value added concentrating among larger, more capital-intensive establishments (Babu, 2026). Our capital-wedge finding is the allocative counterpart of that diagnosis: the factor whose shadow prices are furthest apart is the one the growth pattern has been leaning on hardest.

One of the levers below is not hypothetical. Bau & Matray (2022) show that opening an industry to foreign equity compresses the capital wedge precisely where local banking is weakest, and raises the treated industries’ Solow residual. Capital-market deepening is therefore a policy with demonstrated traction on the margin our decomposition says is the largest, and the states our panel identifies as most capital-distorted are the natural place to apply it.

The policy reading follows from the decompositions rather than from the headline gains. The headline (121–184 per cent) is an upper bound on a reform no instrument delivers — complete within-industry wedge elimination. The attainable and targetable margins are smaller and specific: capital-market deepening where the capital wedge is widest (the eastern states); competition and pass-through policy, which the planner counterfactual scores as the single most pro-labour intervention available (+8.7 points of wage share before any reallocation); and, for the size-dependent agenda that dominates current debate, the recognition that its efficiency content lies at the formality boundary — below this sample — while within the organised sector the size margin is thin and its removal would not pay for its distributional cost. What reform of the organised sector cannot credibly promise, on these estimates, is that improved allocation alone will raise labour’s share of income; without the pricing margin, it lowers it.

8 Conclusion

We measured within-industry misallocation in Indian organised manufacturing over 2012–2022, on a newly built plant panel, inside a demand system that lets markups vary with plant size and respond to reallocation. Misallocation is large. Counterfactual gains from equalising marginal revenue products run from 121 to 184 per cent among the plants the demand system can price, which hold about half of value added, and capital wedges carry them in every year and every state. But the level is not the interesting part. Measured dispersion falls by a fifth over the decade, on a path no single cross-section could reveal, and roughly half of that improvement happened inside surviving plants while the rest is composition, which the endpoint decomposition cannot divide into entry, exit and sample-frame change.

The wedges are also real. A plant’s wedge has a one-year autocorrelation of 0.71 and is still correlated 0.46 after five years, so what we measure is a durable feature of the plant rather than transitory survey noise, although persistent mismeasurement and capital adjustment costs could also produce it. And the wedges are aimed: within industries, revenue-recovered TFPQ and TFPR correlate at 0.70, where an undistorted economy would give zero, a figure that is partly mechanical because both are built from the same revenue. India’s distortions fall hardest on the plants that would use resources best, which is why its measured gains exceed those found for China and for India’s own 2000s.

Three messages emerge, and each is a claim about how this literature should work rather than about India alone.

First, the dispersion of revenue productivity does not tell you the gain, and it does not tell you whether the gain is real. Our markups account for under two per cent of measured dispersion yet remove 7 to 19 per cent of the reallocation dividend, so the mapping from dispersion to gain is not proportional. The mapping from dispersion to *credibility* is looser still. Two economies with identical dispersion can differ in how persistent their wedges are, in how tightly those wedges track productivity, and in how much of the spread is survey error. Those three things are separately measurable on any panel, and reporting a variance without them says less than it appears to.

Second, the accounting choice is worth more than most of the sensitivities the literature runs. Holding sample and technology fixed and varying only whether productivity is computed on value added or on gross output, the within-industry variance moves by a factor of between 6.8 and 8.1. That is larger than the effect of any parameter we vary in Section 6. The canonical framework is value-added throughout, so a large part of the headline magnitude reported for India is a consequence of subtracting a materials bill worth 70 to 90 per cent of revenue. The trend survives the choice. The level does not.

Third, efficiency and distribution come apart, and the wedge between them

runs through market power. Equalising marginal products lowers the aggregate labour share by 2.65 percentage points once markups are allowed to re-optimize, against 0.12 points when they are frozen. Disciplining pricing instead raises labour’s share by 8.7 points before any reallocation occurs. For a labour-abundant economy this is the result with the sharpest policy content, and it is invisible to any framework that treats a markup as a tax. Pairing our state panel with the companion paper’s monopsony geography sharpens it further: employer wage-setting power tracks the labour wedge and not the capital wedge, so the part of misallocation that bears on workers is not the part that dominates the aggregate.

None of this says the reallocation agenda is misconceived. It says that the agenda has been argued on a number, the headline gain, that is the least robust thing the accounting produces, while the robust things — the factor anatomy, the trend, the persistence, the incidence — have been reported less often and carry the policy content. On Indian data over this decade, the binding allocative problem is capital, it is durable, it is aimed at the productive, and removing it would not by itself raise labour’s share of income.

Ten limitations bound the claims. (i) The framework prices only within-industry misallocation among surviving organised-sector plants; cross-industry wedges, entry, and the unorganised sector — where allocation problems are plausibly worse — are outside it. (ii) All counterfactuals are static: transition costs, dynamic innovation effects, and the adjustment-cost component of wedges (Asker et al., 2014) are unmodelled, and part of measured dispersion is noise (Bils et al., 2021), making the gains upper bounds twice over — though Section 5.3 shows the noise share is the smaller part, since between two-fifths and three-quarters of wedge variance is a permanent feature of the plant. (iii) The value-added technology with external capital shares is maintained, not tested; Section 6 quantifies the sensitivity and explains why the domestic alternatives are worse, but the dependence remains. (iv) Value added excludes energy and service inputs, which the panel does not carry. Energy intensity varies within four-digit industries, so the omission does not cancel in the within-cell comparisons the framework makes and inflates measured dispersion by an amount we cannot sign. The point carries a particular sting: Section 5.3 cites Allcott et al. (2016) on Indian electricity shortages as a leading example of exactly the durable, location-attached friction the persistence results pick up, and electricity is the input the measure omits. Some of the persistent capital wedge attributed to credit and land may be unmeasured energy cost. (v) The state analysis rests on **twenty-one** states. The across-state spread and the monopsony join are estimated on that many observations, the partial correlations of Section 5.11 more so, and none of them should be read as precise. (vi) Markups here are demand-side objects; their agreement with the production-side estimates is reassuring but certifies neither.

(vii) The labour-share counterfactual holds sectoral factor supplies fixed and prices only the product-market wedge; general-equilibrium wage responses and the markdown

channel measured in the companion paper are beyond the model. (viii) The 2019 boundary case shows the model class has a solvable domain, and the bootstrap shows how ragged that domain is: about one replication in seven fails to solve, and the failures are concentrated among draws whose calibrated elasticity is high, so the reported intervals are conditional on convergence and too narrow at the upper end (Section 6.4). High-elasticity calibrations anywhere in this literature deserve corresponding caution. (ix) The superelasticity is identified from how markups vary with plant size, and [Ren & Zhang \(2025\)](#) show that monopsony can generate the incomplete pass-through usually attributed to demand curvature. Since the companion paper documents monopsony in a large minority of these plants, part of our calibrated curvature may be a wage response rather than a demand property. Separating them needs a labour-supply curve estimated jointly with demand, which is beyond this design and is the most important extension we can identify. (x) The counterfactual gains exclude the plants the Kimball solver cannot price, which are few but large: 39 to 51 per cent of value added in each year (Table 9). The gains are those of the remaining half of the sector.

Within those bounds, the paper’s central finding stands: in Indian manufacturing, market power determines how much of measured misallocation is real, how much of the promised gains are attainable, and — most consequentially for a labour-abundant economy — how much of the reallocated output markups retain.

References

- Allcott, H., Collard-Wexler, A., & O’Connell, S. D. (2016). How do electricity shortages affect industry? Evidence from India. *American Economic Review*, 106(3), 587–624.
- Asker, J., Collard-Wexler, A., & De Loecker, J. (2014). Dynamic inputs and resource (mis)allocation. *Journal of Political Economy*, 122(5), 1013–1063.
- Atkeson, A., & Burstein, A. (2008). Pricing-to-market, trade costs, and international relative prices. *American Economic Review*, 98(5), 1998–2031.
- Azar, J., & Marinescu, I. (2024). Monopsony power in the labor market: from theory to policy. *Annual Review of Economics*, 16, 491–518.
- Babu, M. S. (2026). Recent manufacturing growth in India. *Economic and Political Weekly*, 61(16). India’s Industrial Transformation series.
- Banerjee, A. V., & Duflo, E. (2005). Growth theory through the lens of development economics. In P. Aghion & S. N. Durlauf (Eds.), *Handbook of Economic Growth* (Vol. 1A, pp. 473–552). Amsterdam: Elsevier.
- Baqee, D. R., & Farhi, E. (2020). Productivity and misallocation in general equilibrium. *Quarterly Journal of Economics*, 135(1), 105–163.

- Bartelsman, E., Haltiwanger, J., & Scarpetta, S. (2013). Cross-country differences in productivity: The role of allocation and selection. *American Economic Review*, 103(1), 305–334.
- Basole, A., & Narayan, A. (2020). Long-run performance of the organised manufacturing sector in India: An analysis of sub-periods and industry-level trends. *Economic and Political Weekly*, 55(10), 35–44.
- Bau, N., & Matray, A. (2022). *Misallocation and capital market integration: evidence from India*. Working paper, June 2022.
- Bergquist, L. F., Lashkari, D., & Verhoogen, E. (2026). Wedges: A microeconomic perspective on misallocation. In *Handbook of Development Economics* (Vol. 6, forthcoming). Elsevier.
- Bertrand, M., Hsieh, C.-T., & Tsivanidis, N. (2021). Contract labor and firm growth in India. NBER Working Paper 29151.
- Besley, T., & Burgess, R. (2004). Can labor regulation hinder economic performance? Evidence from India. *Quarterly Journal of Economics*, 119(1), 91–134.
- Bils, M., Klenow, P. J., & Ruane, C. (2021). Misallocation or mismeasurement? *Journal of Monetary Economics*, 124, S39–S56.
- Bishnoi, A., Dash, R. K., Garg, B., & Sahoo, P. (2024). Boosting the labour-intensive manufacturing in India. *Economic and Political Weekly*, 59(34). India’s Industrial Transformation series.
- Bollard, A., Klenow, P. J., & Sharma, G. (2013). India’s mysterious manufacturing miracle. *Review of Economic Dynamics*, 16(1), 59–85.
- Brandt, L., Tombe, T., & Zhu, X. (2013). Factor market distortions across time, space and sectors in China. *Review of Economic Dynamics*, 16(1), 39–58.
- Brooks, W., Kaboski, J., Li, Y. A., & Qian, W. (2021). Exploitation of labor? Classical monopsony power and labor’s share. *Journal of Development Economics*, 150, 102627.
- Buera, F. J., Kaboski, J. P., & Shin, Y. (2011). Finance and development: A tale of two sectors. *American Economic Review*, 101(5), 1964–2002.
- Bun, M. J. G., & de Winter, J. (2019). *Measuring trends and persistence in capital and labor misallocation*. DNB Working Paper 639. Amsterdam: De Nederlandsche Bank.
- Carrillo, P., Donaldson, D., Pomeranz, D., & Singhal, M. (2023). *Misallocation in firm production: A nonparametric analysis using procurement lotteries* [Working paper].
- Chari, V. V., Kehoe, P. J., & McGrattan, E. R. (2007). Business cycle accounting. *Econometrica*, 75(3), 781–836.

- Das, S. K., & Pant, M. (2006). Measuring market imperfection in the manufacturing sector: theory and evidence from India. *Journal of International Trade & Economic Development*, 15(1), 63–79.
- David, J. M., & Venkateswaran, V. (2019). The sources of capital misallocation. *American Economic Review*, 109(7), 2531–2567.
- De Loecker, J., & Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6), 2437–2471.
- De Ridder, M., Grassi, B., & Morzenti, G. (2024). *The hitchhiker’s guide to markup estimation: assessing estimates from financial data*. Working paper, February 2024.
- Dotsey, M., & King, R. G. (2005). Implications of state-dependent pricing for dynamic macroeconomic models. *Journal of Monetary Economics*, 52(1), 213–242.
- Duranton, G., Ghani, E., Goswami, A. G., & Kerr, W. R. (2015). *The misallocation of land and other factors of production in India* (Policy Research Working Paper 7221). World Bank.
- Edmond, C., Midrigan, V., & Xu, D. Y. (2015). Competition, markups, and the gains from international trade. *American Economic Review*, 105(10), 3183–3221.
- Edmond, C., Midrigan, V., & Xu, D. Y. (2023). How costly are markups? *Journal of Political Economy*, 131(7), 1619–1675.
- Feenstra, R. C. (2003). A homothetic utility function for monopolistic competition models, without constant price elasticity. *Economics Letters*, 78(1), 79–86.
- Foster, L., Haltiwanger, J., & Syverson, C. (2008). Reallocation, firm turnover, and efficiency: Selection on productivity or profitability? *American Economic Review*, 98(1), 394–425.
- García-Santana, M., & Pijoan-Mas, J. (2014). The reservation laws in India and the misallocation of production factors. *Journal of Monetary Economics*, 66, 193–209.
- Ghatak, M., & Mookherjee, D. (2025). Misallocation and development: A review. Working paper.
- Gopinath, G., Kalemli-Özcan, S., Karabarbounis, L., & Villegas-Sanchez, C. (2017). Capital allocation and productivity in South Europe. *Quarterly Journal of Economics*, 132(4), 1915–1967.
- Gupta, A. (2022). *Demand for quality, variable markups and misallocation: evidence from India*. Working paper, Dartmouth College.
- Haltiwanger, J., Kulick, R., & Syverson, C. (2018). *Misallocation measures: the distortion that ate the residual*. NBER Working Paper 24199.

- Holguín, J., Ocampo, S., & Salgado, S. (2026). *Markup accounting*. Working paper, University of Western Ontario and Wharton.
- Hopenhayn, H. A. (2014). Firms, misallocation, and aggregate productivity: a review. *Annual Review of Economics*, 6, 735–770.
- Hopenhayn, H., & Rogerson, R. (1993). Job turnover and policy evaluation: A general equilibrium analysis. *Journal of Political Economy*, 101(5), 915–938.
- Hsieh, C.-T., & Klenow, P. J. (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics*, 124(4), 1403–1448.
- Hsieh, C.-T., Klenow, P. J., Kyei Manu, E., & Rockall, E. (2026). Misallocation versus inequality. Working paper.
- Kabiraj, S. (2020). Analyzing the sources of misallocation in Indian manufacturing: a gross-output approach. *Asian Development Review*, 37(1), 67–92.
- Kimball, M. S. (1995). The quantitative analytics of the basic neomonetarist model. *Journal of Money, Credit, and Banking*, 27(4), 1241–1277.
- Klenow, P. J., & Willis, J. L. (2016). Real rigidities and nominal price changes. *Economica*, 83(331), 443–472.
- Liang, Y. (2023). Misallocation and markups: Evidence from Indian manufacturing. *Review of Economic Dynamics*, 51, 161–176.
- Li, N., & Wang, X. (2021). Estimating resource misallocation: distinguishing factor market distortions from variable markups. *Economics Letters*, 209, 110118.
- Midrigan, V., & Xu, D. Y. (2014). Finance and misallocation: Evidence from plant-level data. *American Economic Review*, 104(2), 422–458.
- Mohammad, A., Sandoz, C., & Sodsriwiboon, P. (2021). *Resource misallocation in India: the role of cross-state labor market reform*. IMF Working Paper WP/21/51. Washington, DC: International Monetary Fund.
- NITI Aayog (2026). *Key sectors to position India as a global manufacturing hub*. New Delhi: Government of India.
- Olley, G. S., & Pakes, A. (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica*, 64(6), 1263–1297.
- Peter, A., & Ruane, C. (2026). Intermediate input substitutability. *Econometrica* (forthcoming).
- Peters, M. (2020). Heterogeneous markups, growth, and endogenous misallocation. *Econometrica*, 88(5), 2037–2073.

- Petrin, A., & Levinsohn, J. (2012). Measuring aggregate productivity growth using plant-level data. *RAND Journal of Economics*, 43(4), 705–725.
- Press Information Bureau (2025). *Production Linked Incentive schemes: investment, production, exports and employment*. Government of India, Ministry of Commerce and Industry.
- Ren, K., & Zhang, D. R. (2025). *Price markups or wage markdowns?* Working paper, September 2025.
- Reserve Bank of India (2024). *Measuring productivity at the industry level: the India KLEMS database — data manual 2024*. Mumbai: Reserve Bank of India.
- Restuccia, D., & Rogerson, R. (2008). Policy distortions and aggregate productivity with heterogeneous establishments. *Review of Economic Dynamics*, 11(4), 707–720.
- Restuccia, D., & Rogerson, R. (2017). The causes and costs of misallocation. *Journal of Economic Perspectives*, 31(3), 151–174.
- Shukla, R. (2026, June 19). Productivity up, wages flat: how India’s manufacturing growth has stopped working for its workers. *Ideas for India*.
- Trottner, F. (2022). *Misallocations in monopsonistic labor markets*. Working paper, University of California, San Diego.
- United Nations (2015). *Transforming our world: the 2030 Agenda for Sustainable Development*. Resolution A/RES/70/1. New York: United Nations General Assembly.
- Weinberger, A. (2020). Markups and misallocation with evidence from exchange rate shocks. *Journal of Development Economics*, 146, 102494.
- Zhang, X., & Xia, J. (2022). *Misallocation under heterogeneous markups and non-constant returns to scale*. Working paper, Tilburg University and Peking University.

A Sample construction and deviations from the reference implementation

The input panel is the companion paper’s build (its Appendix A documents the raw block files, fixed-width layouts, identifier structure, deflation, and verification suite); this framework consumes the nominal-value subset. The cleaning chain, in order: (1) drop plant-years with non-positive value added, output, capital, employment, wage bill, or materials; (2) retain NIC-4×year cells with at least fifty plants; (3) recover μ, τ_K, τ_Y , TFPR, TFPQ per (8)–(10); (4) trim the 1 per cent tails of log TFPR and log TFPQ deviations by year. From 481,322 input plant-years this yields 429,534.

Three unavoidable deviations from Liang (2023), each with its expected direction of effect. (i) **Capital** is the closing rather than the average of opening and closing book value (opening values are not carried consistently across the ASI’s format regimes); this adds noise, and a level difference for fast-growing plants, to τ_K and TFPR, but no evident bias to their dispersion. (ii) **Value added** excludes energy and service inputs, which were not extracted into this panel. This overstates the level of value added, and the usual defence — that the framework compares only within an industry-year — covers anything common to the cell but not this. Energy intensity varies *across plants inside* a four-digit industry-year, with vintage, product mix, captive generation and location, so a plant whose energy bill is unmeasured carries overstated value added and overstated revenue productivity, and enters the dispersion as a wedge. The direction is signed but the magnitude is not, and it is discussed as a limitation in Section 8 rather than left here. (iii) **No profit-based filter** is applied (the panel carries no profit variable); her drops of non-positive-profit observations are subsumed partly by the non-positive-value-added drop and the trims. A fourth, deliberate choice: the tobacco division inherits $\alpha_s = 0.95$ from the U.S. mapping and is retained in the baseline but examined separately (Section 6.7).

B Numerical implementation

The counterfactual equilibria are computed with Liang (2023)’s replication code executed verbatim (Stata/Mata): for each industry, the demand index solves the aggregator identity (13) by secant iterations, nested inside the outer secant loop that moves ς to hit the elasticity target (16); convergence tolerance and iteration caps are hers. One engineering addition was required by the larger industry count of the modern NIC-4 structure: industries whose demand-index iteration fails to bracket a root at extreme parameter values are solved on a widened bracket with the same tolerance, and any industry that still fails is reported (none does in any baseline year; the capped-sector count is zero in every row of Table 20). The 2017 gate — reproducing the reference-year solution from a cold start before the full run — and the per-year logs are part of the project’s verification suite. The HKKR diagnostics (do-files 36–37) implement (21) and the size-neutralising counterfactual at $\sigma = 3$ with harmonic VA-share-weighted $\overline{\text{TFPR}}_s$, HK’s equations 15–16 conventions.

C The full state panel

Table 22 reports all states and union territories with at least 2,000 estimation plant-years, ranked by TFPR dispersion.

Table 22: Misallocation by state, pooled 2012–2022: within-industry variance ($\times 100$) of log TFPR, MRPL, MRPK. Ranked by var TFPR.

State	N	var TFPR	var MRPL	var MRPK	Region
Odisha	6,368	77.6	63.9	156.5	East
Andhra Pradesh	16,095	76.5	62.9	161.2	South
West Bengal	17,300	75.9	67.4	158.0	East
Kerala	12,940	72.9	57.9	147.9	South
Himachal Pradesh	7,193	72.0	70.0	125.3	North
Jharkhand	5,089	71.7	72.3	141.3	East
Telangana	13,210	71.6	59.9	137.5	South
Puducherry	2,190	70.1	64.4	132.8	South
Jammu & Kashmir	3,651	68.6	64.7	127.1	North
Madhya Pradesh	13,630	67.9	61.4	129.4	Central
Maharashtra	47,511	66.9	58.2	131.4	West
Karnataka	24,168	65.4	56.8	132.9	South
Tripura	3,549	64.5	38.1	131.6	Northeast
Goa	2,480	64.3	65.5	108.3	West
Dadra & Nagar Haveli	2,157	64.2	58.6	119.7	West
Rajasthan	20,143	63.8	52.7	128.8	North
Delhi	7,172	63.0	48.7	138.2	North
Uttarakhand	10,011	62.9	63.5	115.0	North
Uttar Pradesh	37,692	62.8	51.0	134.5	North
Chhattisgarh	6,635	60.6	61.8	119.3	Central
Tamil Nadu	54,801	60.4	51.9	128.7	South
Bihar	8,931	59.1	37.0	125.8	East
Daman & Diu	3,670	57.0	46.8	110.6	West
Gujarat	41,896	56.5	51.5	118.3	West
Assam	10,977	56.2	49.6	104.7	Northeast
Haryana	24,381	49.4	41.8	109.3	North
Punjab	19,081	49.1	37.0	110.2	North

D Gross-output versus value-added accounting: detail

Both measures in (22) are computed on the same plant-years from the same estimated technology — the companion paper’s translog gross-output elasticities, averaged within NIC-2 and CRS-normalised into shares (s_K, s_L, s_M) , with the value-added shares $a_K = s_K/(s_K + s_L)$ — so the columns of Table 23 differ *only* in accounting. Dispersion is the within-(NIC-4×year) standard deviation of the log measure.

Table 23: Within-industry variance ($\times 100$) of TFPR under gross-output and value-added accounting, same sample, same technology.

Year	N	Gross-output	Value-added	Ratio VA/G
2012	33,130	20.2	136.8	6.8
2013	36,332	19.3	134.0	7.0
2014	38,370	16.9	130.9	7.8
2015	37,322	16.9	132.5	7.8
2016	39,087	17.3	132.5	7.7
2017	39,212	16.6	122.8	7.4
2018	39,402	14.7	115.6	7.8
2019	39,738	15.6	113.5	7.3
2020	40,256	15.7	115.6	7.4
2021	40,899	14.2	114.1	8.1
2022	39,072	14.9	107.1	7.2

Two lessons. The value-added measure is seven to eight times more dispersed in every year: subtracting materials — 70–90 per cent of revenue for most plants — mechanically levers up any noise or heterogeneity in the materials share into the productivity residual. And both columns fall over the decade in parallel, so the paper’s trend statements are invariant to the accounting; only levels are at stake, and levels are not what the paper’s conclusions ride on.